

ICA Lens: Interpreting Language Models Without Training Another Dictionary

Sida Liu^{1†} Feijiang Han^{2†} ABC² DEF¹

¹Independent Researcher ²University of Maryland

Finding interpretable directions in language-model activations is a central goal of mechanistic interpretability. Sparse autoencoders (SAEs) have become the dominant tool for this purpose, but their modern workflow often turns interpretation into a training pipeline of its own. For each model, layer, activation site, and sparsity regime, one may need to train, tune, store, and evaluate a large overcomplete dictionary before analysis can begin. This cost limits rapid exploration and motivates a simpler question: how much useful structure can be recovered before training another neural dictionary? We revisit independent component analysis (ICA), a classical method for finding non-Gaussian directions, as a compact lens for language-model interpretability. Prior SAE work has often treated ICA as a weak baseline. We find that this weakness is not inherent to ICA alone, but largely reflects a mismatch between standard ICA recipes and modern LLM activations—high-dimensional residual streams contain large-norm token outliers and slow-converging components that can destabilize fitting. We address these issues with a practical GPU-parallel FastICA pipeline, activation normalization, and auditable convergence diagnostics, enabling efficient layer-wise analysis across modern LLMs. Across GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B, we find that ICA recovers compact components with substantially more exceptional projection statistics than random directions and public SAE decoder directions. These components span a spectrum from token-local patterns to broader contextual signals. Through manual audits and case studies, we show that they expose human-readable structure in residual streams and token embeddings, including lexical form, local syntax, discourse context, semantic ambiguity, and embedding-level lexical organization. We also find that ICA and SAE provide complementary views of activation space. ICA recovers both SAE-aligned axes and interpretable directions that are weakly covered by nearby SAE features, while ICA directions more often trace broader context-sensitive patterns than SAE features. We further evaluate these directions through targeted intervention and sparse probing benchmarks, where ICA remains competitive in the small-to-medium feature-budget regime without per-layer gradient-based dictionary training. These results suggest that ICA should not be dismissed as a failed SAE baseline, but used as a fast, compact, and complementary first lens for exploring language-model representations.

1. Introduction Sida

Understanding language models requires more than measuring their input–output behavior: we would like to disentangle their internal activations into directions that correspond to recognizable mechanisms, concepts, contexts, or computations. Such directions should be readable enough for human analysis and useful enough to support probing, comparison, and hypothesis testing. Sparse autoencoders (SAEs) have become the dominant approach to this feature-discovery problem (????). They learn an overcomplete dictionary whose sparse latents reconstruct model activations, giving researchers a large set of candidate feature directions to inspect. SAE variants differ in architecture, sparsity mechanism, dictionary size, and training recipe, and recent training-light alternatives such as ITDA reduce some engineering burden. But these methods largely share the same source of signal: reconstruction error under a sparsity constraint. They ask whether activations can be rebuilt from a small number of learned directions.

†Corresponding author(s): Sida Liu me@liusida.com, Feijiang Han fejhan@umd.edu

In this paper, we study a different source of signal for feature discovery: non-Gaussian activation statistics. A direction that responds selectively to a token form, phrase template, syntactic construction, semantic context, or discourse state should not behave like an average random projection over many token positions; it may activate rarely, prefer one sign, separate recurring modes, or respond strongly only in specific contexts. This intuition is supported by our observation that public SAE decoder directions are themselves substantially more non-Gaussian than random directions, even though SAEs are trained for sparse reconstruction rather than non-Gaussianity. We therefore ask: if we optimize directly for this statistical signature, without reconstruction-sparse training, how much interpretable and useful structure can we recover?

Independent component analysis (ICA) is a classical method that directly optimizes this inductive bias. After centering and whitening, ICA searches for a compact linear basis whose one-dimensional projections are maximally non-Gaussian. It therefore differs from SAE-style methods at the level of the discovery signal itself. ICA does not learn directions by minimizing reconstruction error. It searches activation space for statistically exceptional directions. If these directions are interpretable and behaviorally useful, then some feature-like structure in LLM activations is already visible in activation geometry before training another dictionary.

Standard ICA, however, is not reliable out of the box on modern LLM activations. Residual streams are high-dimensional, contain large-norm token outliers, and include slow or oscillating components that can dominate convergence statistics. A naive FastICA run can be slow, unstable, or reject a layer even when many components have already stabilized. We therefore build a practical ICA pipeline for LLM activations, combining GPU-parallel FastICA, row-normalized preprocessing, robust convergence diagnostics, and adaptive refitting for difficult layers. These changes make ICA stable enough for layer-wise analysis across GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base.

Using this pipeline, we find that ICA recovers compact directions that are statistically significant, human-readable, and useful for analysis. ICA components have substantially higher excess kurtosis than random projections and public SAE decoder directions. To understand what this statistical exceptionality corresponds to in text, we introduce effective receptive field (ERF), which measures how much left context is needed to recover a component’s signed response. ERF shows that ICA components span a spectrum from token-local patterns to broader contextual signals, and provides one possible explanation for the observed link between kurtosis and interpretability. In human audits and case studies, ICA components expose surface form, word categories, phrase templates, sentence-level constructions, global topics, long-range repetition, semantic ambiguity, discourse context, and embedding-level lexical organization.

We also compare ICA with SAE features. The two decompositions partially overlap, showing that non-Gaussianity and sparse reconstruction can recover related axes of activation space. But they do not coincide: many interpretable ICA components are only weakly covered by their nearest public SAE feature, and their activation patterns often trace broader context-sensitive signals rather than localized sparse events. Under SAEbench sparse probing, ICA remains competitive with high-capacity public SAEs in compact feature-budget regimes and outperforms PCA, ITDA, and compact Matryoshka SAE variants in our evaluated settings.

These results reposition ICA as a statistics-first lens for LLM interpretability. ICA is not a replacement for overcomplete SAEs, which remain better suited for building large high-resolution feature dictionaries. Its role is different: before training or browsing a large sparse-reconstruction dictionary, ICA can cheaply expose a compact set of non-Gaussian directions that are small enough for manual inspection yet rich enough to reveal meaningful structure in model internals. Our main claim is therefore not merely that ICA is cheaper than SAEs. It is that reconstruction error is not the only useful signal for feature discovery: non-Gaussianity

alone can recover interpretable, context-sensitive, partly SAE-aligned, partly distinct, and behaviorally useful directions in LLM activation space.

2. Introduction

Understanding the internal activations of language models is a central goal of mechanistic interpretability. Sparse autoencoders have become the dominant approach to this problem by decomposing activations into sparse latents over an overcomplete dictionary of learned directions (????). However, interpreting language models with SAEs is increasingly becoming a training problem of its own. For each model, layer, activation site, and sparsity regime, one may need to train, tune, store, and evaluate a separate dictionary before analysis can begin. Gemma Scope, for example, required hundreds of SAEs, tens of millions of learned latents, 4–16B training tokens per SAE, roughly 20 PiB of saved activations, and over 20% of GPT-3’s training compute (?). Such scale is valuable when the goal is a comprehensive feature vocabulary, but it is heavy when the goal is rapid exploration or small-budget intervention. We therefore need a compact first lens for understanding activation structure before training or browsing a large SAE dictionary.

Intuitively, interpretable directions are selective rather than typical random projections over all tokens. They may activate on a small set of contexts, separate recurring modes, prefer one sign, or respond to lexical, syntactic, semantic, or discourse patterns. Such selectivity should leave a statistical footprint in activation space. We study non-Gaussianity as a simple and measurable footprint of this kind. This is motivated by our observation that sparse reconstruction implicitly favors statistically exceptional axes in activation space: public SAE decoder directions are substantially more non-Gaussian than random directions, despite being trained for sparse reconstruction rather than non-Gaussianity. A natural alternative is therefore to explicitly use non-Gaussianity as an inductive bias for finding interpretable directions more efficiently.

Independent component analysis (ICA) gives a compact implementation of this idea. After centering and whitening, ICA finds a linear basis whose one-dimensional projections are as non-Gaussian as possible, returning a small set of statistically exceptional directions.

However, ICA has not become a widely used first lens for LLM activation analysis. We find that this is not due to a lack of promise in the idea itself, but to two practical and evaluative gaps in applying ICA to LLM activations. First, standard ICA does not work reliably out of the box on modern LLM activations. Residual streams are high-dimensional, contain large-norm token outliers, and include a small number of slow or oscillating components that can dominate convergence. As a result, a naive CPU FastICA run can be slow, fail to converge on many layers, and return few or no usable components. Although prior work treated ICA as an automated-interpretability baseline (?), its implementation used a plain CPU FastICA pipeline without optimization.¹ We therefore argue that it does not fully reflect the potential of ICA when adapted to LLM activations. Second, subsequent evaluation and analysis work on activation interpretation has largely used PCA, rather than ICA, as the main classical baseline (?). Prior evaluations and analyses have not systematically tested ICA through large-scale human component annotation, comparison with SAE directions, or downstream measures of feature usefulness.

We therefore revisit ICA in this paper to explore and evaluate its potential as an interpretability lens for LLM activations more comprehensively.

Making ICA practical for LLM activations. We first address the practical gap by building an optimized ICA pipeline for LLM activations. The pipeline makes FastICA GPU-parallel and adapts it to the geometry

¹https://github.com/HoagyC/sparse_coding/blob/main/autoencoders/ica.py

of LLM residual streams through activation normalization, robust convergence diagnostics, and adaptive component-count selection. These recipes turn vanilla ICA from a brittle baseline into a stable and auditable pipeline for layer-wise LLM activation analysis. For example, on GPT-2 Small with 1M activations, our optimized pipeline increases the number of accepted layers by 400.0% and reduces the total number of FastICA iterations by 21.5%.

Evaluating and analyzing ICA directions. We then address the evaluation and analysis gap by asking three questions: whether ICA directions are interpretable to human analysts, how they differ from SAE directions, and whether they are useful beyond description. We first build an interactive ICA explorer and manually annotate ICA components at scale across GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base. Through manual audits and case studies, we show that ICA components expose human-readable structure in both residual streams and token embeddings. The recovered components include lexical form, local syntax, phrase templates, discourse context, semantic ambiguity, long-range repetition, and embedding-level lexical organization. We also find that ICA components gradually shift from token-local patterns toward broader context-dependent signals in deeper layers.

To understand what ICA recovers and how it relates to SAEs, we compare the two decompositions from statistical, geometric, and activation-pattern perspectives. These findings suggest that ICA and SAE provide complementary views of activation space rather than one being a subset of the other. First, fitted ICA components have substantially more exceptional non-Gaussianity than random directions and public SAE decoder directions. Components with stronger non-Gaussianity also tend to require less context to activate, which helps explain why they are easier to interpret. Second, ICA components partially overlap with SAE decoder directions, but many interpretable ICA components are only weakly covered by their nearest SAE features. At the activation level, ICA directions often trace broader context-sensitive signals, whereas SAE features more often fire as localized sparse events. This suggests that non-Gaussianity maximization and sparse reconstruction recover related but non-identical axes of activation space.

To test whether the recovered directions are useful beyond human description, we use SAE Bench sparse probing to evaluate whether ICA directions carry concept-relevant information (?). We compare ICA with public SAEs with high-capacity overcomplete dictionaries, Matryoshka SAE variants with smaller dictionary sizes (?), ITDA as a training-light sparse-coding alternative (?), and PCA as a classical statistical baseline. Across models, ICA remains competitive with public SAEs and consistently outperforms PCA, ITDA, and Matryoshka SAE variants. These results show that searching for highly non-Gaussian directions is more informative than searching for high-variance directions, and that ICA is a strong compact baseline for concept detection among non-SAE methods.

ICA should therefore not be dismissed as a failed SAE baseline. It is a fast, compact, and complementary first lens for exploring language-model representations before training or browsing a large SAE dictionary.

3. ICALen: Making FastICA Practical for LLM Activations

We summarize the standard parallel FastICA pipeline in Appendix A. The algorithm is conceptually simple, but a direct application to modern LLM activations is brittle. LLM activation matrices are high-dimensional, contain large-norm outlier token positions, and often have a small set of slow or oscillating components that dominate convergence statistics. Thus, a naive FastICA run can reject an entire layer even when most directions have already stabilized and remain useful for analysis.

We make FastICA practical with three simple recipes. First, row normalization reduces the effect of activation-norm outliers before whitening and improves the fitting dynamics (Section 3.2). Second, when the

strict max-LIM criterion fails, we soften the acceptance rule to p95-LIM, so a small tail of difficult directions does not decide whether the layer is usable (Section 3.3). Third, for layers that remain difficult, we use an adaptive refitting procedure that reduces the target component count only when needed, giving each layer the highest accepted resolution under the same convergence standard (Section 3.4). These recipes turn FastICA from a fragile baseline into a reliable pipeline for producing stable, analyzable directions.

3.1. Activation Corpus

We evaluate ICA on GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base. For each model, we fit ICA independently at the embedding layer and at each residual-stream layer. The embedding layer provides a non-contextual view of token-identity structure, while residual-stream layers expose contextual representations formed during autoregressive processing.

For residual-stream fitting, we collect hidden states from one million randomly sampled token positions from the Pile-10k training split for each model. Documents are truncated to context length 1024, and token positions are sampled uniformly without replacement using a fixed random seed. We run the model on the truncated documents and record hidden states at the sampled positions, so each activation is computed with its proper left context rather than as an isolated token.

Let $x_i^{(\ell)} \in \mathbb{R}^d$ denote the residual-stream activation at sampled token position i and layer ℓ . For each layer, we form

$$X^{(\ell)} = \begin{bmatrix} x_1^{(\ell)}, \dots, x_n^{(\ell)} \end{bmatrix}^\top \in \mathbb{R}^{n \times d}, \quad n = 10^6.$$

This gives a task-agnostic activation corpus that is not tuned to any later case study or benchmark. For the embedding-layer analysis, we replace the contextual activation matrix with the model’s static input embedding matrix and pass it through the same fitting interface. The embedding matrices contain 50,257 rows for GPT-2, 256,000 rows for Gemma 2 2B, and 248,320 rows for Qwen 3.5 2B Base.

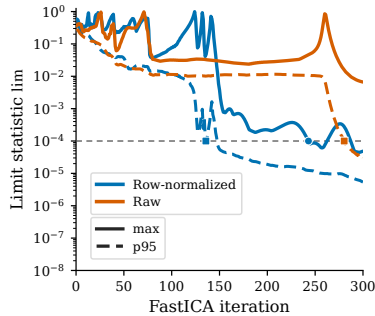


Figure 1: Layer-3 convergence diagnostic for GPT-2 Small. Solid curves show max-LIM, and dashed curves show p95-LIM. Row normalization lowers the fitting statistics, while p95-LIM separates typical convergence from worst-case outlier behavior.

Rows	Max-LIM only		Max-LIM + p95 fallback	
	Raw	Norm.	Raw	Norm.
1k	2 / 3,431	9 / 2,177	2 / 3,431	9 / 2,177
3k	3 / 3,294	5 / 3,155	3 / 3,294	5 / 3,155
10k	2 / 3,499	6 / 3,206	4 / 3,383	6 / 3,206
100k	7 / 3,224	7 / 3,055	9 / 3,010	9 / 2,710
1M	2 / 3,492	8 / 3,107	6 / 2,954	10 / 2,741

Table 1: GPT-2 Small convergence yield. Each cell reports accepted layers / total iterations across 12 residual-stream layers. For the p95 fallback, we use max-LIM whenever it succeeds and p95-LIM only when the strict criterion fails. Bold marks the best setting for each row.

3.2. Row-Normalized ICA

The first practical difficulty is the input geometry seen by the optimizer. When FastICA is applied directly to raw LLM activations, we often observe slow or unstable convergence. Prior work suggests two relevant

intuitions. First, raw LLM activations can contain large-norm outlier token positions (???). Second, many activation-space interventions operate on directions rather than raw vector norms (??). We therefore normalize activations before fitting ICA, which makes the fitting dynamics more stable and efficient.

Concretely, before the usual centering and whitening, each activation vector is normalized by its ℓ_2 norm:

$$r(x_i) = \frac{x_i}{\max(\|x_i\|_2, \epsilon)},$$

where ϵ is a small numerical constant. ICA is then fit on the row-normalized matrix $\tilde{X}$, whose rows are $r(x_i)^\top$.

Figure 1 shows a representative example on GPT-2 Small layer 3. The normalized curves stay below the raw-activation curves through most of training. The normalized max-LIM curve reaches the 10^{-4} threshold within the iteration budget, while the raw max-LIM curve remains above it. Table 1 shows the same pattern at the layer level. Under the strict max-LIM criterion, normalization improves the accepted-layer count from 2 to 9 at 1k fitting rows, from 2 to 6 at 10k rows, and from 2 to 8 at 1M rows. It also usually reduces the total number of fitting iterations.

3.3. Robust Convergence Acceptance

The second practical difficulty is convergence diagnosis. Standard FastICA uses a worst-case stopping rule:

$$\max_j \text{LIM}_j < \tau.$$

This rule accepts a fit only when every component has stabilized up to sign. We keep this strict criterion whenever it succeeds. However, in high-dimensional LLM activation fits, a small number of slow or oscillating components can dominate the maximum and cause the whole layer to be rejected. For exploratory interpretability, this is too conservative: many stable directions may be discarded because of a small unstable tail.

To make convergence less sensitive to the single hardest component, we record the full distribution of per-component LIM values. When the strict rule fails within the iteration budget, we use a softer fallback criterion:

$$p_{95}(\text{LIM}_j) \leq \tau.$$

This accepts a layer when almost all components have stabilized, while still flagging the remaining tail as difficult.

Figure 1 illustrates this distinction. The dashed p95-LIM curves decrease more smoothly, while the solid max-LIM curves show sharper spikes and slower worst-case behavior. The aggregate effect appears in Table 1. With 1M fitting rows, raw activations accept only 2 of 12 GPT-2 layers under max-LIM alone, but 6 layers with the p95 fallback. With row normalization, the same comparison improves from 8 to 10 accepted layers and reduces total iterations from 3,107 to 2,741.

3.4. Adaptive Refit for Difficult Layers

Some layers still fail to meet the acceptance criterion within the iteration budget even after row normalization. We handle these cases with an adaptive refit procedure. We start with $m = d$ and halve m until the layer satisfies either the strict max-LIM criterion or the p95-LIM criterion, or until a minimum component count

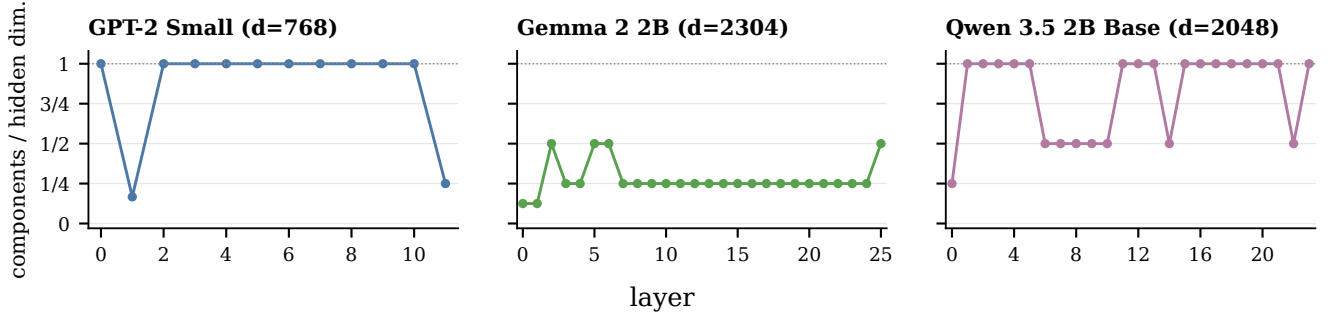


Figure 2: Final ICA component counts selected by the adaptive refit procedure. For each layer, we initially attempt to fit the full hidden dimension and reduce the target component count when convergence fails. Dashed horizontal lines indicate the hidden dimension of each model.

$m_{\min} = 16$ is reached. This gives each layer the highest accepted resolution under the same convergence standard.

Figure 2 summarizes the final accepted component counts across layers. Many layers converge at the full hidden dimension, while some layers require reduced component counts. The accepted component count is therefore also a diagnostic of fitting difficulty. A layer that accepts $m = d$ has a full-dimensional ICA basis. A layer that only accepts $m < d$ still provides a valid set of non-Gaussian directions.

4. Statistical Structure of ICA Directions

The previous section **Modified: made ICA stable and efficient enough to fit on LLM activations**. We now ask what kind of directions it recovers. A useful direction should pass two checks. First, it should be statistically exceptional rather than an arbitrary rotation of the activation space. Second, its exceptionality should correspond to recoverable structure in text, ranging from token-local patterns to broader contextual signals.

This section separates these two steps. We first verify that ICA components are substantially more non-Gaussian than random projections and SAE decoder directions. We then introduce **Modified: effective receptive field (ERF)**, a **diagnostic** for each component. ERF measures how much context is sufficient to recover a component’s activation at a target token, helping us characterize the context dependence of ICA components across layers and models.

4.1. Non-Gaussianity Separates ICA Directions from Random Projections

For a direction v and a set of row-normalized activations $\{\tilde{h}_i\}_{i=1}^n$, we measure the excess kurtosis of its scalar projection

$$z_i(v) = \langle v, \tilde{h}_i \rangle$$

as

$$\kappa(v) = \frac{\frac{1}{n} \sum_{i=1}^n (z_i(v) - \bar{z}(v))^4}{\left(\frac{1}{n} \sum_{i=1}^n (z_i(v) - \bar{z}(v))^2\right)^2} - 3.$$

A Gaussian projection has excess kurtosis close to zero. Larger values indicate that the direction has a heavier-tailed or otherwise more exceptional projection distribution.

We compare three families of directions: random unit directions as a null baseline, SAE decoder directions **Modified: as learned sparse-dictionary directions**, and ICA score directions recovered by our fitted ICA maps. **Modified: All three direction families are evaluated as projection directions on the same row-normalized activation distribution. This comparison therefore tests whether a direction is statistically exceptional under a common input geometry.** Figure 3 shows a consistent separation across all three model families. ICA directions are substantially more non-Gaussian than random directions and SAE decoder directions across layers, confirming that our fitted ICA components successfully recover the statistical objective of the method on LLM activations.

Interestingly, SAEs are not trained to maximize kurtosis, but their decoder directions already show elevated non-Gaussianity, suggesting that non-Gaussianity is a common statistical signature of many **Modified: learned feature directions**. This gives us a useful perspective on why ICA can serve as an interpretability lens: ICA makes this inductive bias explicit by directly searching for directions with exceptional projection statistics, rather than learning it implicitly through sparse reconstruction as in SAEs. In Sections 5–6, we show that these non-Gaussian directions from ICA are also human-interpretable and behaviorally useful under downstream feature evaluations.

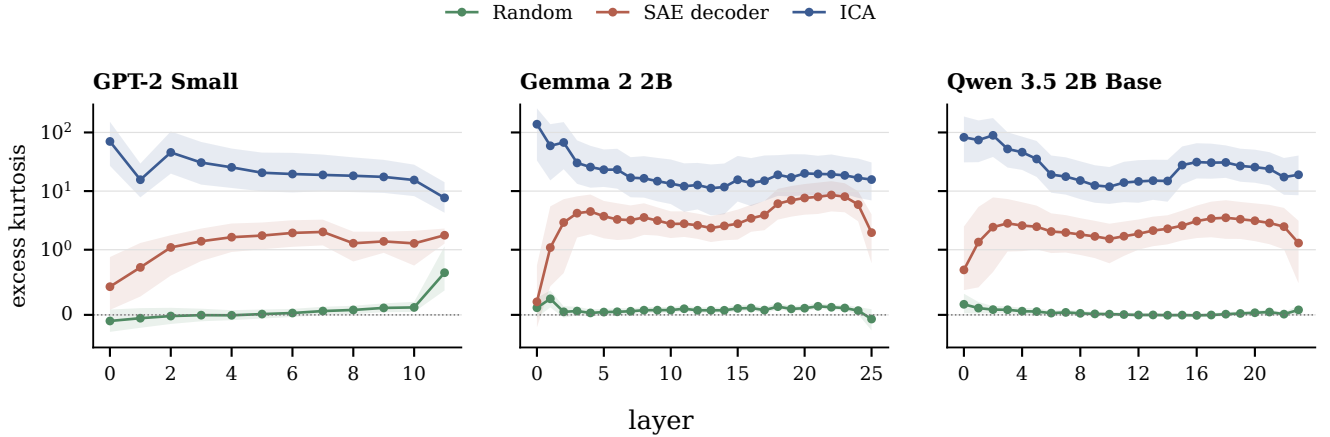


Figure 3: ICA recovers highly non-Gaussian directions in LLM activation space. For each model and layer, we project row-normalized residual-stream activations onto random unit directions, public SAE decoder directions, and fitted ICA score directions, then compute the excess kurtosis of each projection distribution. Across GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base, random projections remain close to Gaussian, SAE directions show elevated non-Gaussianity, and ICA directions are the most non-Gaussian, validating the statistical objective of the ICA fit. Statistics are computed over one million token positions from Pile-10k; random baselines use 32,768 Gaussian unit directions per layer. Lines show medians across directions, and shaded bands show interquartile ranges.

4.2. From Non-Gaussianity to Context Dependence

Non-Gaussianity tells us that a direction is statistically unusual, but not what textual structure makes it unusual. A high-kurtosis component may correspond to a local token pattern, a short phrase, or a broader contextual signal. We therefore need a way to ask whether ICA is only finding local detectors or also recovering components whose activation depends on context. To separate these cases, we introduce **Modified: effective receptive field (ERF)**.

ERF asks a simple question: how much left context is sufficient to recover the same signed component response at a target token? Small-ERF components can be recovered from the token itself or a short suffix,

while large-ERF components require a broader preceding context.

Computing ERF. ERF is computed from a set of evidence examples for each component. For a target token position t in a sequence $x_{1:T}$, let $h_t(x_{1:t})$ be the residual-stream activation computed with the full available context, and let $s_j(h_t)$ be the signed score of ICA component j . We first identify the highest-scoring example of component j in absolute value and record its sign. We then collect all examples with the same sign whose absolute score exceeds half of this maximum absolute score, forming an evidence set $\mathcal{E}_j$. In practice, this typically yields around 20 evidence examples per component.

For each evidence example (x, t) , we construct suffixes ending at the target token with lengths

$$x_{t-k+1:t}, \quad k = 1, 2, \dots, K_{\max}.$$

Let $h_t^{(k)}$ denote the final-token activation obtained from the length- k suffix. We then recompute ICA scores at this token. A suffix of length k is said to recover component j if the component remains among the top 15 components ranked by absolute score and preserves its original sign. Formally, let

$$R_j(x, t, k) = \mathbb{1} \left[j \in \text{Top}_{15} \left(\{|s_r(h_t^{(k)})|\}_r \right) \wedge \text{sign}(s_j(h_t^{(k)})) = \text{sign}(s_j(h_t(x_{1:t}))) \right].$$

Modified: We search suffix lengths in increasing order and stop at the first successful recovery. The sample-level ERF is the shortest suffix length that recovers the component,

$$\text{erf}_j(x, t) = \min\{k \in \{1, \dots, K_{\max}\} : R_j(x, t, k) = 1\}.$$

If no suffix of length at most $K_{\max}$ succeeds, we assign $\text{erf}_j(x, t) = K_{\max}$ and mark the example as unrecovered within the tested window. In our experiments, $K_{\max} = 11$. The component-level ERF² is the mean sample-level ERF over evidence examples:

$$\text{ERF}(j) = \frac{1}{|\mathcal{E}_j|} \sum_{(x,t) \in \mathcal{E}_j} \text{erf}_j(x, t).$$

ERF reveals a local-to-contextual spectrum. **Figure 4 Modified: summarizes component context dependence as a layerwise distribution, which uncovers many insights.** First, ICA components span a broad local-to-contextual spectrum. Many components are recovered from the target token or a short suffix, but a substantial fraction require longer context. Thus, ICA is not merely finding lexical detectors; it also exposes directions whose activation depends on the surrounding text.

Modified: Second, ERF provides a quantitative lens on layer specialization. It shows that component scope changes with depth through a gradual shift in mixture proportions, not through a sharp handoff between layer types. Early layers are dominated by token-local components but already contain context-dependent directions, while later layers contain more medium- and long-context components yet still retain many token-local directions. Thus, layers are better described by the mixture of component types they contain than by a single local-to-global label. The distribution also shows a non-monotonic pattern. The largest-ERF components, including those recovered only at the maximum tested length or not recovered within the 11-token window, are often most prevalent in middle layers. This makes middle layers a useful target for inspecting context-dependent ICA components.

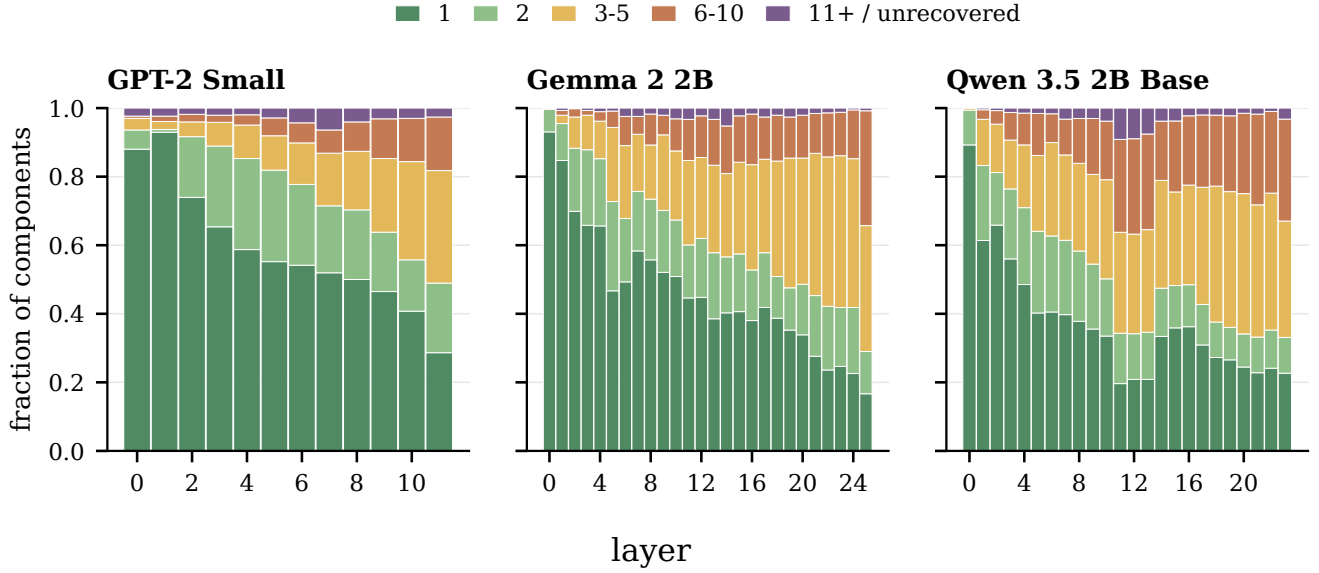


Figure 4: Layerwise distribution of effective receptive field (ERF). For each ICA component, we average the sample-level ERF over its evidence examples and bucket components by the resulting mean ERF. Each stacked bar shows the fraction of components in one layer. The final bin includes components **Modified: not recovered within the 11-token window, reported as 11+.**

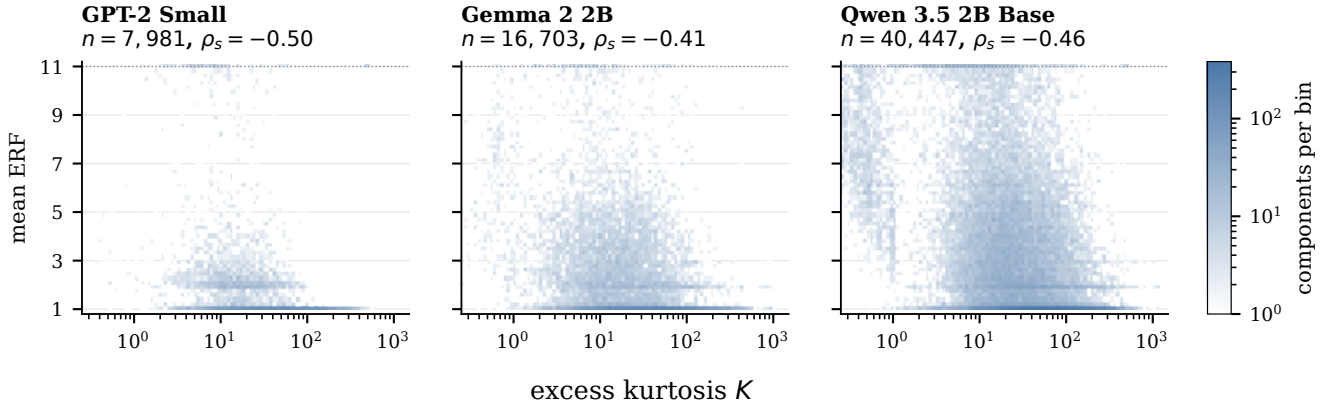


Figure 5: Relationship between effective receptive field (ERF) and excess kurtosis. Each cell counts ICA components with a given excess-kurtosis range and mean ERF. Across all three models, components with larger excess kurtosis tend to have smaller ERFs, with Spearman correlations ranging from -0.41 to -0.50 . This indicates that high-kurtosis components are typically more local, while broad-context components tend to have lower kurtosis and larger ERFs.

ERF offers one explanation for the kurtosis–interpretability link. **Modified: Prior SAE work found that feature kurtosis is positively correlated with automated interpretability scores (?), suggesting that more kurtotic activation distributions tend to be easier for existing auto-interpretation pipelines to explain. However, the correlation itself does not identify the underlying reason. Figure 5 shows a consistent negative**

²The half-maximum evidence threshold, Top-15 recovery criterion, and $K_{\max}$ are heuristic choices. We use the same protocol for all models and layers, and treat ERF as a comparative diagnostic rather than an intrinsic property of a component.

relationship between ERF and excess kurtosis across all three models. Components with larger excess kurtosis tend to have smaller ERFs. These components are often activated by narrow, recurring patterns such as tokens, surface forms, or short phrases, so their top examples expose the triggering pattern directly and are easier to summarize for both human annotators and LLM-based automatic interpretation methods. By contrast, larger-ERF components depend on broader context. Their activation conditions may not be visible at the target token or in a short local window, making them less sharply kurtotic and harder to explain from standard top-example views. ERF therefore gives an operational interpretation of the kurtosis signal: high kurtosis is useful partly because it correlates with locality, and locality makes components easier to explain.

Modified: This interpretation also makes ERF useful for organizing annotation. Small-ERF components can often be analyzed from top examples alone, while larger-ERF components typically require context ablations or longer evidence windows. ERF therefore serves as a proxy for annotation difficulty. We use this ERF-guided strategy in our manual analysis in Section 5.2, and it may also help guide future automatic annotation systems.

5. Human Interpretation of ICA Components

After quantifying the statistical structure and context dependence of ICA components, we now ask a more direct interpretability question: can these components be inspected, labeled, and reused in a controlled way? This section describes the inspection interface and annotation protocol, audits randomly sampled components, and then uses the resulting labels to analyze contextual and lexical structure.³

5.1. An Interactive Lens for Component Inspection

To support component interpretation, we built an interactive explorer that keeps the relevant evidence for each component visible during analysis. Given an input sequence, the interface displays the strongest ICA components at each token, ranked by absolute signed score. For each selected component, it shows the signed score, current working label, top-scoring examples, original document context, ERF, excess kurtosis, and annotation metadata. The explorer also links components across adjacent layers using cosine similarity between ICA directions. These links often provide useful clues when related patterns persist across depth.

The explorer is a quality-control tool for annotation. Its main role is to prevent labels from being assigned from a few isolated top examples without checking whether the component behaves consistently under additional contexts. The interface therefore supports both passive inspection of high-scoring examples and active hypothesis testing with targeted prompts.

5.2. Annotation Protocol

Given an activation h_t , we rank ICA components by absolute signed score $|s_j(h_t)|$. The magnitude indicates how salient component j is at the target token, while the sign selects one side of the ICA direction. Since ICA directions are identifiable only up to sign, annotations are sign-specific. The two sides of the same component may receive different labels when they correspond to different activation patterns.

Our annotation protocol is evidence-controlled and ERF-guided. For each component side, the expert annotator first checks the component’s ERF to estimate the expected scope of evidence. Low-ERF components

³We treat component labels as working hypotheses: short descriptions of the dominant pattern captured by a signed ICA direction, not ground-truth semantic definitions.

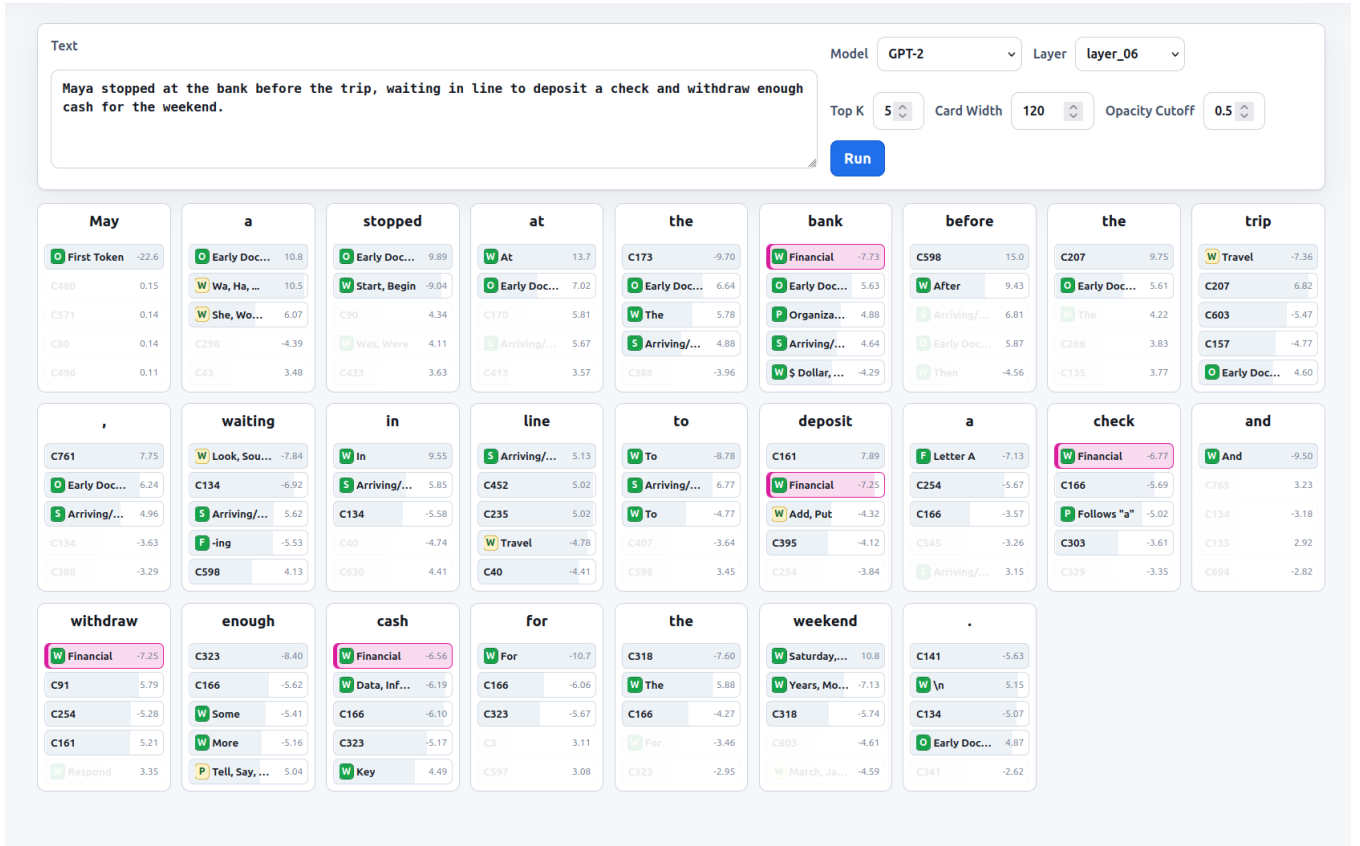


Figure 6: Interactive inspection view for GPT-2 Small layer 6. For each token, the explorer displays the strongest ICA components, their signed scores, and their working labels. The interface keeps component scores, evidence examples, ERF, kurtosis, and annotation metadata visible during annotation, enabling consistent inspection across GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base.

can often be interpreted from the target token and its local context, while high-ERF components usually require broader document context or explicit context ablations. The annotator then inspects a fixed set of high-scoring examples, including the target token, signed component score, and surrounding text. When the pattern is ambiguous, the annotator also examines middle-range, near-zero, and opposite-sign examples to avoid explaining only the most extreme activations.

The annotation proceeds as an iterative hypothesis-testing process. The annotator first proposes a concise working label from the evidence examples. They then construct targeted prompts that should activate the component and matched counterexamples that should not. The explorer is used to check whether the component appears at the expected token positions with the expected sign. For context-dependent components, the annotator additionally performs context ablations by progressively removing preceding text and monitoring whether the component remains active. The label is revised until it is stable across top examples, targeted tests, and, when needed, context ablations.

Each labeled side receives a working label, a confidence level, and an interpretation type. Confidence is one of high, medium, low, or unclear. A high-confidence label consistently explains the observed behavior; a medium-confidence label captures a clear but harder-to-summarize pattern; a low-confidence label gives only a partial explanation; and an unclear label fails to provide a reliable interpretation. The interpretation type records the scale of the pattern: form components respond to surface features such as capitalization or

initial letters, word components are tied mainly to token identity, phrase components respond to short local constructions, sentence components track sentence-level structure or topics, long-range components depend on earlier distant words or repetitions, global components reflect broader genre or discourse setting, position components respond mainly to token position, and sophisticated components appear to implement more conditional patterns.

5.3. Random Component Audit

To assess whether ICA components are practically inspectable, we conduct a random component audit. For each model, we uniformly sample 50 converged ICA components without replacement from the full component inventory, including embedding-layer components. For each sampled component, the expert annotators label the dominant signed side—the sign of the strongest absolute-score example. Empty labels and placeholder labels such as “?” are counted as unclear.

Model	High	Med.	Low	Unc.	W	P	F	S	G/L/Pos/X
GPT-2 Small	44	4	1	1	23	8	7	5	6
Gemma 2 2B	43	3	2	2	15	11	3	7	13
Qwen 3.5 2B Base	40	1	4	5	16	14	1	8	6
Total	127	8	7	8	54	33	11	20	25

Table 2: Random component annotation summary. We sample 50 ICA components per model and annotate the dominant signed side of each component. High/Med./Low/Unc. report annotation confidence. Type columns count the main interpretation type: word (W), phrase (P), form (F), sentence (S), and the union of global, long-range context, position, and sophisticated components (G/L/Pos/X). The complete annotation table appears in Appendix 7.

Table 2 shows that most sampled components admit usable working labels under this protocol. Across 150 randomly sampled components, 142 receive non-unclear labels and 127 receive high-confidence labels. The labels are not limited to token identity. They include local form patterns, word categories, phrase templates, sentence-level constructions, global topics, position effects, and long-range or conditional patterns. We interpret these results as evidence of practical inspectability, because a randomly sampled ICA component often supports a concise and testable human hypothesis.

Table 3 shows representative prompt-level cases from the initial audit and how a working label is tested in practice. For each component, label-consistent prompts are paired with matched label-inconsistent prompts. We report the signed component score and its rank among components at the target token.

The examples illustrate several levels of component behavior. Some components behave like local token or form detectors, such as the After component. Others track structured local contexts, such as citation-final punctuation. Global and long-range components are more demanding: the gaming component activates on discourse from game play rather than on a single lexical item, while the section-header component depends on repeated structure across a longer context. This also supports the role of ERF as a guide for annotation difficulty. Table 4 gives a compact view of representative audited labels across models.

We next ask whether the labels remain credible under review by another experts. For each of the 127 high-confidence labels, a second expert auditor inspected the component evidence and the initial working label. The auditor assigned a 0–10 score and wrote a revised label when the original description could be made more precise. This secondary audit serves as a consistency check: it asks whether the first-stage expert labels remain supported under an independent expert inspection of the same component evidence.

Component	Expected	Prompt	Score	Rank
L0/C192: After high, Word, negative side, ERF 1.0, $\kappa = 298$				
L0/C192	activate	I went outside to play <u>after</u> I finished my homework.	-22.447	1
L0/C192	activate	The puppy ran <u>after</u> the ball.	-22.700	1
L0/C192	not activate	I went outside to play <u>when</u> I finished my homework.	-1.017	18
L0/C192	not activate	The puppy ran <u>through</u> the ball.	+1.503	21
L7/C17: Scientific research / citation high, Sentence, negative side, ERF 2.0, $\kappa = 33$				
L7/C17	activate	Smith, J., & Lee, K. (2021).	-9.480	1
L7/C17	activate	Vaswani, A., et al. (2017). Attention is all you need.	-16.003	1
L7/C17	not activate	Smith and Lee (2021) is a fictional patent citation used here for illustration purposes only.	-2.051	14
L7/C17	not activate	In 2017, Ashish Vaswani led the team that published Attention Is All You Need, introducing the Transformer model.	+0.114	639
L10/C368: Gaming language high, Global, positive side, ERF 3.2, $\kappa = 31$				
L10/C368	activate	Don't fight before dragon unless our jungler has Smite up; save your ult for <u>their</u> engage.	+21.492	1
L10/C368	activate	If the tank does not kite during the second phase, the healers get overwhelmed <u>by</u> the damage.	+16.429	1
L10/C368	not activate	Don't leave before dinner unless everyone has arrived; save your announcement for <u>their</u> arrival.	+0.158	634
L10/C368	not activate	If the manager does not delegate during the busiest part of the project, the team gets overwhelmed <u>by</u> the workload.	-0.328	471
L8/C738: Repetition of a prior section header medium, Long-range, negative side, ERF 7.7, $\kappa = 23$				
L8/C738	activate	### Section 1: Discovery ... ### Section <u>2</u> : Analysis	-11.160	1
L8/C738	activate	Cat: maomao. Dog: wowowow? <u>C</u> at: mamoa.	-7.496	2
L8/C738	not activate	### Discovery ... ### <u>A</u> alysis	+0.352	502
L8/C738	not activate	At 3:30 p.m., the race began, and the score was <u>3</u> :2 after the first round.	-1.737	55

Table 3: Representative prompt-level cases from the random component audit. Label-consistent prompts usually produce a large signed score and high rank at the target token, while matched label-inconsistent prompts reduce the score or rank. These controlled cases complement top-example inspection and make the working labels easier to falsify.

Model	Component	ERF	Type	Working label
GPT-2 Small	L0/C192	1.0	Word	After
GPT-2 Small	L9/C445	5.3	Long-range	Refer to a spokesperson
Gemma 2 2B	L13/C30	8.6	Sophisticated	Refer back to one of two alternatives
Gemma 2 2B	L10/C261	9.2	Long-range	Repetition and number increases
Qwen 3.5 2B Base	L18/C1794	3.5	Global	Study protocol
Qwen 3.5 2B Base	L22/C511	4.9	Sentence	Either/or, whether/or construction

Table 4: Representative labels from the random component audit. The examples span token-local, sentence-level, global, long-range, and conditional patterns. Labels are working descriptions under the inspection protocol, not ground-truth semantic definitions.

The secondary audit supports the high-confidence subset. Among the 127 audited labels, 121 are supported and the remaining 6 are partially supported; none are rejected. Moreover, 112 of the 127 labels receive a score of at least 8. The audit also shows that many initial labels can be sharpened, for example from “Gaming language” to “video-game discourse” or from “Scientific research, Citation” to “scientific reference-title final period.”

Table 5 shows representative rows from the secondary expert audit across all three models. Each row includes the initial label, the audit score, and the revised label suggested after the second inspection. The full audit table is provided in Appendix ??.

Model	Component	Initial label	Score	Revised label
GPT-2 Small	L0/C192	After	10	after token
GPT-2 Small	L7/C17	Scientific research, Citation	9	scientific reference-title final period
GPT-2 Small	L10/C368	Gaming language	9	video-game discourse
GPT-2 Small	L9/C445	Refer to a spokesperson	8	spokesperson/spokesman name reference
Gemma 2 2B	L24/C355	Turkish language	10	Turkish-language token/subword
Gemma 2 2B	L13/C30	Refer back to one of two recently introduced alternatives	9	reference to one member of a recent pair
Gemma 2 2B	L10/C261	Repetition and number increases	8	repeated identifier with increasing numeric suffix
Gemma 2 2B	L20/C263	Cryptography, Computer Security	8	cryptography / security context
Qwen 3.5 2B Base	L18/C1794	Study protocol	10	clinical study protocol/approval language
Qwen 3.5 2B Base	L19/C824	Follow, violate rules	10	rule compliance/violation language
Qwen 3.5 2B Base	L22/C511	Either or, Whether or, One way or another	9	alternative/concessive construction
Qwen 3.5 2B Base	L11/C1211	Because of, As a result of	9	causal preposition phrase

Table 5: Representative rows from the secondary expert audit of high-confidence labels. The table reports the initial working label assigned in the first-stage expert annotation, the 0–10 audit score, and the revised label suggested by the second expert auditor. The revised labels usually preserve the original interpretation while making it more specific.

5.4. Case Studies: Reading Component Mixtures

The random audit asks whether ICA components can be labeled in isolation. We now ask how these labels behave when they are placed back onto concrete inputs. The case studies below use three complementary views. The first fixes a polysemous token and follows how its representation decomposes across layers. The second fixes two labeled components and traces their scores across sentences. The third removes context entirely and studies how embedding-layer ICA decomposes familiar lexical relations.

5.4.1. Contextual Decomposition of a Polysemous Word

Polysemous words provide a controlled setting for studying contextual decomposition because the surface token is fixed while the surrounding evidence changes. We use a short paragraph containing four occurrences of bank. Two refer to a financial institution, denoted F1 and F2, and two refer to a river edge, denoted R1 and R2.

Maya stopped at the bank [F1] before the trip, waiting in line to deposit a check and withdraw enough cash for the weekend. The teller asked about her travel plans, and Maya said she was driving north to

visit the old village by the river. By sunset she was sitting on the grassy bank [R1] with her shoes beside her, watching the water curl around stones and reeds. It amused her that the same word had followed her all day: first a bank [F2] with counters, accounts, and vaults, then a bank [R2] of earth holding the river in place.

Figure 7 is a component-level decomposition of four occurrences of the same token. Each column corresponds to one bank token in the paragraph. Each row corresponds to a layer. Within each cell, we list the five ICA components with largest absolute score at that token position. The bar length shows relative score magnitude within the cell. The text label gives our working interpretation of the signed component side, and the badge records annotation type and confidence. Thus, the figure is not a word-sense classifier. It shows which interpretable directions are active in the model state when GPT-2 processes that token under its available left context.

The embedding row provides a useful control. All four columns are identical because the input embedding depends only on the token identity bank. The strongest embedding components include Bank, Mortgage, A Room or Place, Group, Team, and Letter B. Before contextual processing begins, all four occurrences share the same lexical representation.

The residual-stream layers show how this shared lexical state separates under context. F1 occurs after the prefix “Maya stopped at the.” The later evidence, such as “deposit a check” and “withdraw enough cash,” is unavailable at the target token, yet the prefix already supports a financial reading. Later layers therefore contain components related to finance, organizations, companies, investment, and departments. This shows that the model can begin to disambiguate bank from left context alone.

R1 has a different mixture because the prefix already points to a physical scene. The paragraph has mentioned a village by the river, and the immediate prefix is “sitting on the grassy.” The active components shift toward labels such as Water, Place Name, Building, Place, Crouch, Sit, Body Movement, and Area, Field. These labels should be read as scene factors rather than a single river-bank feature. The token representation combines water, location, posture, movement, and outdoor space.

F2 returns to the financial sense, but its context differs from F1. The immediate prefix is “first a,” and the preceding paragraph has already introduced both senses of bank. The model therefore represents F2 as part of a repeated-word contrast while also activating financial components such as Financial, Business, Organization, Pay, Department, and Investment. The same cell also contains phrase-level components such as Follows “a”. This illustrates that one token representation can mix semantic content, local syntax, and discourse structure.

R2 is the easiest column to misread. A human reader identifies it as a river-edge use after seeing “bank of earth holding the river in place.” At the target token, however, the model has access only to the prefix “first a bank with counters, accounts, and vaults, then a.” This prefix explicitly mentions the financial sense and sets up a contrast. The finance-related components in R2, including Financial, Department, and Investment, are therefore expected under autoregressive conditioning. The cell reflects the model state at the target token, not the interpretation available after reading the following words.

This token-centered view gives two main lessons. First, ICA does not compress polysemy into one “financial versus river” direction. Each occurrence is represented by a changing mixture of lexical, phrase-level, semantic, positional, and longer-context components. Second, the mixture must be interpreted relative to the information available at the target position. The embedding row is token-driven, while later residual-stream layers incorporate the available prefix and discourse context.

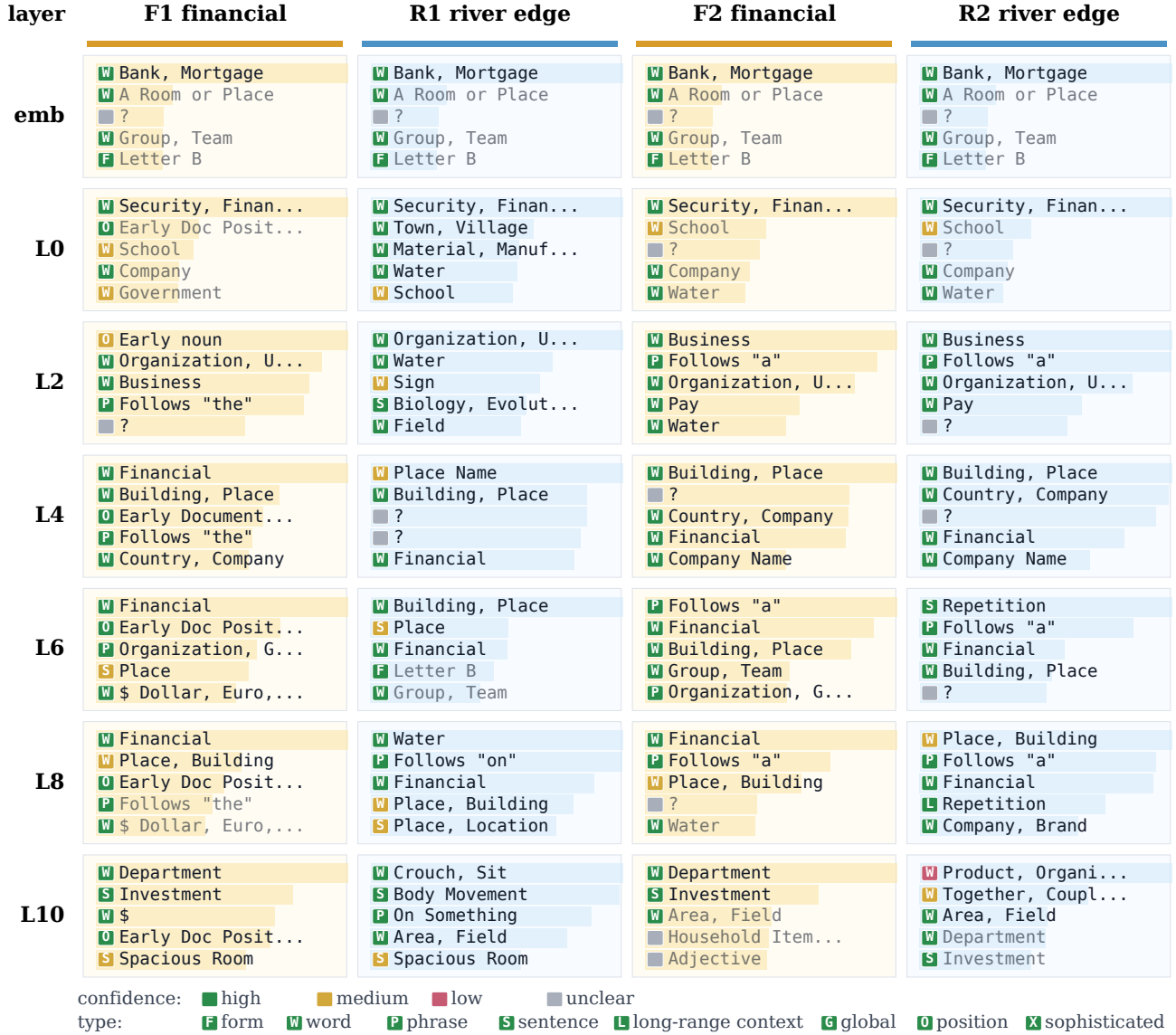


Figure 7: Contextual decomposition of a polysemous word in GPT-2 Small. We probe four occurrences of bank in one paragraph, including two financial-bank uses (F1, F2) and two river-edge uses (R1, R2). Each cell lists the five largest-absolute-score ICA components at the target token in a given layer. Bar lengths show relative score magnitude within the cell, and the legend decodes annotation type and confidence. Across layers, the same surface token is represented by different mixtures of lexical, syntactic, semantic, positional, and longer-context components.

5.4.2. Component Traces Across Sentences

The previous analysis fixes a token and asks how its representation decomposes across layers. We now take the complementary view. We fix two labeled components and ask how each component behaves across sentence positions. This is necessary because a component that appears in the top list for one token may not be a detector for that token alone. It may track a broader context that persists across several related tokens.

Sentence 1. She arrived at the library to study for her exam.

Sentence 2. Maya stopped at the bank before the trip, waiting in line to deposit a check and withdraw enough cash for the weekend.

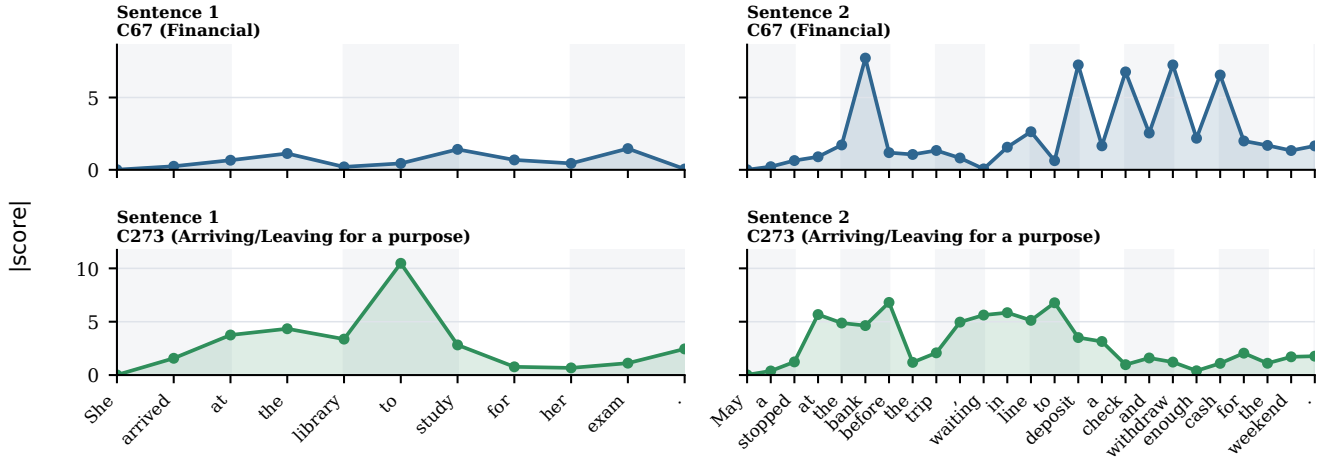


Figure 8: Sentence-level traces for selected GPT-2 Small ICA components. We plot raw absolute ICA scores across two sentences for layer-6 components C67 and C273. C67 tracks a financial-bank context and becomes active across several related tokens in Sentence 2. C273 tracks an arrival- or purpose-related construction and responds strongly to “arrived at the library to study” in Sentence 1, while also responding to related movement and purpose structure in Sentence 2. ICA directions can therefore trace context-sensitive signals across spans, rather than firing only at isolated tokens.

Figure 8 shows this distinction directly. C67 is labeled as a financial component. In Sentence 1, which contains no financial context, its score stays close to zero. In Sentence 2, it becomes active around bank and remains active near deposit, check, withdraw, and cash. If C67 were only a detector for the word bank, we would expect a single spike at that token. Instead, the trace suggests a broader financial-context direction.

C273 illustrates the same point for a construction-level pattern. In Sentence 1, it rises around “arrived at the library to study,” which combines motion, destination, and purpose. The highest response occurs near the to study region, where the purpose relation becomes explicit. In Sentence 2, the same component also responds to related structure, including stopping at a place and waiting in line to perform an action. C273 is therefore better understood as an arrival- or purpose-related direction than as a detector for a single token such as arrived or to.

The insight from this trace view is simple. Some ICA components behave like local token or phrase detectors, but others trace contextual states across a span. Such traces help avoid overly narrow labels. A component whose largest score occurs near one token may still be responding to a larger construction. In our annotation workflow, we use these traces as hypotheses and then check them with targeted prompts and context ablations. Equivalent bank walkthroughs for Gemma 2 2B and Qwen 3.5 2B Base are provided in Appendix ??.

5.4.3. Embedding Components Recover Lexical Structure

The previous two case studies analyzed contextual representations. We now apply the same inspection protocol to token embeddings. This setting is a useful control because token embeddings and residual-stream

activations live in the same d -dimensional representation space, but embeddings are fixed for each token. Embedding-layer ICA therefore asks which non-Gaussian directions organize lexical representations before any context enters the model.

Classical word-embedding analyses often summarize lexical relations through vector offsets, such as $\text{king} - \text{man} + \text{woman} \approx \text{queen}$. This view is useful, but it compresses a relation into one displacement. ICA gives a more inspectable view. Instead of asking whether an analogy can be solved by one offset, we ask which readable directions are active in each token embedding.

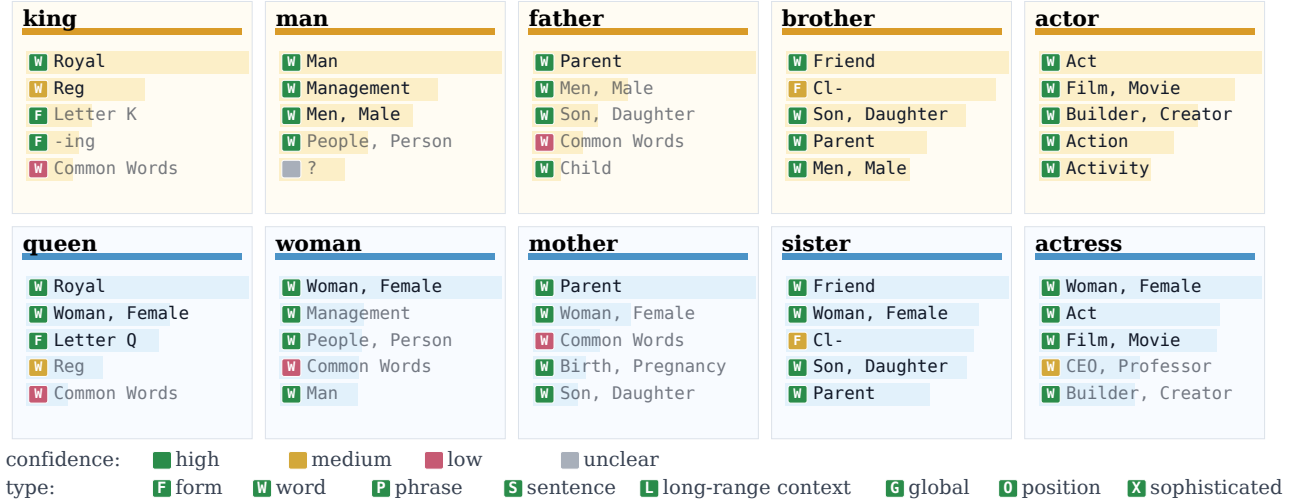


Figure 9: Embedding-layer ICA components for familiar analogy word sets in Qwen 3.5 2B Base. Each box corresponds to one token embedding and lists its five largest-absolute-score ICA components. Columns pair familiar masculine-coded tokens with their feminine-coded counterparts. Bar lengths show relative score magnitude within each box, and the legend decodes annotation type and confidence. The figure shows that familiar analogy pairs are supported by overlapping component directions, including royalty, gender-associated terms, family relations, professions, morphology, and common-token components. GPT-2 Small and Gemma 2 2B examples are shown in Appendix ??.

Figure 9 is a component-level decomposition of static token embeddings. Each box corresponds to one token, and each row inside the box is one high-scoring ICA component for that token embedding.

The first pattern is shared structure with extra factors. For king and queen, both tokens activate a Royal component, which recovers the expected royalty-related direction. Yet the two boxes also contain different additional components. Queen strongly activates Woman, Female and a surface-form component such as Letter Q, while king activates Letter K, morphology, and common-token components. Thus, the familiar pair is not represented only as “royalty plus gender.” It is represented as a mixture of semantic, gender-associated, morphological, and surface-form directions.

The second pattern is that different analogy families decompose in different ways. In the man/woman pair, both tokens activate person-related directions, while gender-associated components distinguish the pair. In the father/mother pair, both tokens share a Parent component and child-relation components. Father is associated with Men, Male, while mother is associated with Woman, Female and Birth, Pregnancy. The brother/sister pair shows a different organization. Both tokens activate directions such as Friend, Son, Daughter, and Parent, suggesting that sibling terms are represented through overlapping social and family-relation components rather than one isolated sibling axis.

The profession pair gives a third view of the same decomposition. Actor is dominated by components related to acting, film, creation, action, and activity. Actress shares several of these directions, including Act, Film, Movie, and Builder, Creator, but also prominently activates Woman, Female. This makes the relation more specific than a profession offset. The embedding contains both the shared occupation-related structure and the gender-associated component that distinguishes the pair.

This component view also makes asymmetries visible. Queen and actress prominently activate components labeled Woman, Female, while king and actor emphasize other directions among their top components. Therefore, ICA can reveal asymmetric associations that are easy to hide when a relation is summarized by a single vector offset.

The main lesson is that embedding-layer ICA turns a coarse analogy view into a component-level explanation. Some directions are shared across a pair, such as royalty, parenthood, or occupation. Others distinguish the pair, such as gender-associated, morphological, profession-related, or common-token components. As in the residual-stream case studies, these labels are working descriptions rather than concept definitions. A component labeled Royal or Woman, Female is useful because it summarizes a recurring direction of variation in the inspected examples. This makes embedding-layer ICA a compact diagnostic for lexical structure before context enters the model.

6. Comparing ICA Components with SAE Features

Sparse autoencoders and ICA both expose directions in activation space, but they are designed around different objectives. SAEs learn large overcomplete dictionaries through reconstruction and sparsity losses. ICA instead returns a compact set of statistically non-Gaussian directions after normalization, centering, and whitening. We therefore should not expect the two decompositions to be identical. The relevant question is more specific: do they recover related structure, and if so, where do they differ?

Modified: We study this question at three levels. We first compare ICA components with public SAE decoder directions geometrically. We then compare their token-wise activation patterns on the same input. Finally, we test whether ICA components can serve as useful feature directions in SAE-oriented SAE Bench evaluations. **Modified: These analyses support a complementary view: ICA overlaps with SAE directions enough to provide convergent evidence, while also exposing compact and context-sensitive directions that are not well described as a small subset of public SAE dictionaries.**

6.1. Directional Overlap

We begin with a geometric comparison. For each ICA component, we find the public SAE decoder direction in the same model and layer with the largest absolute cosine similarity. **Modified: All directions are evaluated as projection directions on the same activation distribution used for ICA analysis, so the comparison measures directional agreement under a common input geometry.** A high nearest-SAE cosine means that an ICA component points in a direction similar to some SAE feature. A low value does not imply that the component is uninterpretable; it only means that the component is not well matched by a single SAE decoder direction. This distinction matters because an overcomplete SAE can distribute related structure across many features, while ICA is constrained to a compact basis.

Figure 10 shows broad nearest-neighbor distributions across all three models. **Modified: Most components fall in a moderate-overlap regime, with a tail of high-overlap components and many weakly matched components. Modified: This rules out two simple interpretations.** ICA is not unrelated to SAE, since many components have a clear SAE neighbor. But ICA is also not merely a compact copy of public SAE dictionaries, since

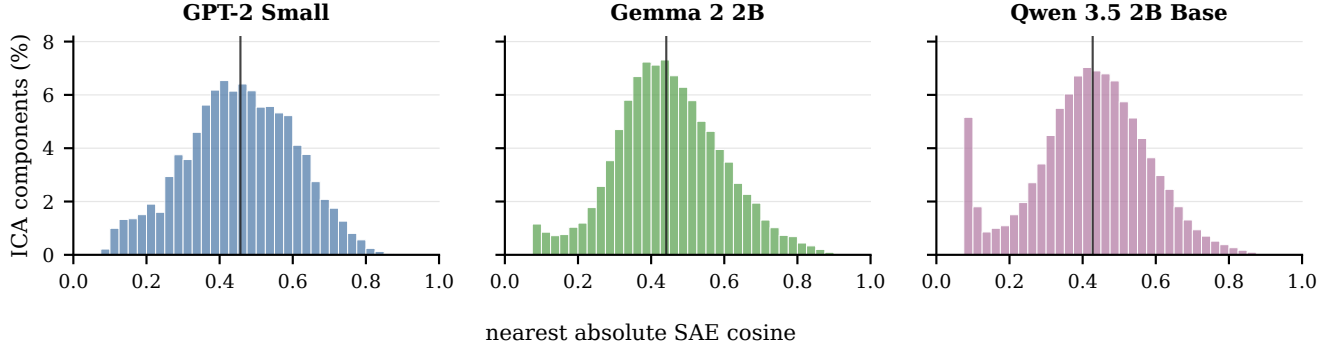


Figure 10: Nearest-SAE overlap for ICA components across models. For each ICA component, we find the public SAE decoder direction in the same layer with maximum absolute cosine similarity. The distributions are broad in GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base. Some ICA components closely align with SAE directions, while many have only moderate or weak nearest-SAE overlap. This partial overlap supports using high-overlap components as convergent evidence while motivating complementary analysis of low-overlap ICA directions.

many components do not align closely with any single SAE decoder direction.

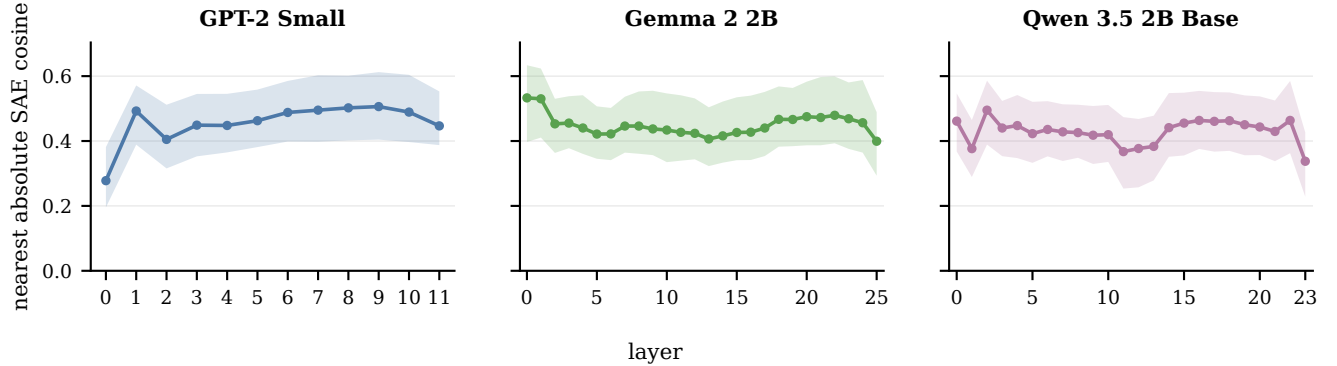


Figure 11: Layer-wise nearest-SAE overlap for ICA components across models. For each layer, points show the median maximum absolute cosine between ICA components and public SAE decoder directions, and shaded bands show interquartile ranges. The partial-overlap pattern is stable across layers in GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base, indicating that ICA-SAE overlap is a broad property of the decompositions rather than a layer-specific artifact.

Figure 11 shows that this pattern is not confined to a few layers. Median nearest-SAE overlap stays in a moderate range across depth, with model-specific variation but no layer where ICA collapses into SAE or becomes orthogonal to it. We use this result as a guide for later analysis. High-overlap components can be treated as convergent evidence that ICA and SAE have found related axes. Low-overlap components are especially interesting when they remain interpretable, because they suggest structure that is visible under the ICA objective but not captured by a single public SAE decoder direction.

Table 6 makes the nearest-SAE comparison concrete. For each model, we select three ICA components from the random-component audit set: the lowest, median, and highest nearest-SAE cosine examples within that sampled set. For each ICA component, we take its strongest absolute-score example, identify the target token for that ICA activation, and then measure whether the nearest SAE feature also activates on that same target-token activation. The “SAE act.” and “SAE rank” columns therefore test functional agreement, not

ICA	Label	Abs-top context	Nearest SAE	cos	SAE act.	SAE rank
GPT-2 low L0/C395	Water, Removed	...im and walk into the kitchen. I've already pulled a bottle of water	F10102	0.10	0	inactive
GPT-2 mid L0/C192	After	...f the tibial tray may remain exposed above the tibial plateau after	F10573	0.44	31.9	1
GPT-2 high L10/C142	Conditional repetition	string "You're pretty good.\$" Route110_Text_16EA2A:: @ 816EA2	F29658	0.75	46.1	14
Gemma low L6/C216	71 in No.	_) ASBCA Nr' 60315)) Under Contract No. HTC71	F9142	0.09	0	inactive
Gemma mid L12/C466	Accusation of bad behavior	...nger the lives of our law enforcement officers to promote a radical	F2168	0.43	9.87	33
Gemma high L24/C510	Female-centered narrative	...he referee, Carlos Ramos have wronged Serena Williams and took from	F9501	0.73	118	3
Qwen low L1/C834	Certain name	6, 1994. Decided July 8, 1994. Rear	F12249	0.08	0	inactive
Qwen mid L18/C1794	?	...ained from the patient included in the study, and the institutional	F3134	0.42	15.1	1
Qwen high L22/C511	?	...trigger, here's a post on how to determine whether they scrolled up	F8570	0.68	4.88	3

Table 6: Sampled ICA components and their nearest public SAE features. Rows are drawn from the preselected random-component audit set, choosing the lowest, median, and highest nearest-SAE cosine example for each model. The context column shows the ICA top-absolute-score example ending at the target token. SAE activation and rank measure whether the nearest SAE feature activates on that same target-token activation.

only geometric agreement: a high cosine neighbor is more convincing if it also fires on the ICA component’s own top example, while a low-overlap neighbor should often remain inactive. For convenience, the SAE feature identifiers are linked to their corresponding Neuronpedia pages when available. This is what we see in the table: low-overlap examples such as GPT-2 L0/C395 have nearest SAE features that are inactive on the ICA top example, whereas mid- and high-overlap examples often activate the nearest SAE feature at the same token.

The GPT-2 high-overlap case illustrates why this comparison is useful. GPT-2 L10/C142 is labeled by our audit as conditional repetition, and its nearest SAE feature is F29658 with cosine similarity 0.75. Neuronpedia labels F29658 as “numerical values and various symbols that appear frequently in mathematical or programming contexts.” However, when we test the ICA top-style sentence, “The event will be organised by the Civil Aviation Authority of Zimbabwe (CAAZ),” the SAE feature peaks on the same token suggested by the ICA component, with activation 8.94. Since this example is not mathematical or code-like, the matched activation suggests that the SAE direction may share the ICA component’s operational behavior more closely than its public label suggests.

The Gemma high-overlap case shows the same caution from another angle. Gemma L24/C510 is labeled in our audit as female-centered narrative, while Neuronpedia labels its nearest SAE feature F9501 as “positive affirmations and expressions of encouragement.” In simple controlled prompts, F9501 activates most strongly for “She’s passionate about books, travel, beauty, and all things cheese.” (99.5), less strongly for the male-substituted version (77.0), and still less for “He’s passionate about cars, football, and video games.” (40.75). This pattern is more consistent with a gendered or female-centered narrative direction than with encouragement alone. Overall, the table and these spot checks suggest that nearest-SAE cosine can reveal real functional overlap, but existing SAE labels should be treated as hypotheses rather than ground truth. If we take a look at the nearest SAE feature for Gemma from Neuronpedia: F9501, we see the label on that website is "positive affirmations and expressions of encouragement", but our label is "Female-centered narrative". We don’t trust the SAE label, so construct three test sentences: "She’s passionate about books,

travel, beauty, and all things cheese." Highest activation is 99.5 and "He's passionate about books, travel, beauty, and all things cheese." Highest score is 77.0 and "He's passionate about cars, football, and video games." Highest score is 40.75. So we guess the actual feature is maybe related to our "Female-centered narrative", not as the neuronpedia label says.

6.2. Activation Pattern Comparison

Directional overlap only compares static directions. It does not tell us whether ICA components and SAE features behave similarly on actual text. To compare their token-wise behavior, we use the same sentence for both decompositions and select the strongest directions at each token. For GPT-2 Small, we select the top-1 ICA component by absolute score and the top-1 SAE feature by activation at each token. We then plot the union of all selected directions across the full sentence. Each row is normalized by its own maximum value in the sentence, so the plot shows the shape of each direction's response rather than comparing absolute magnitudes across different directions.

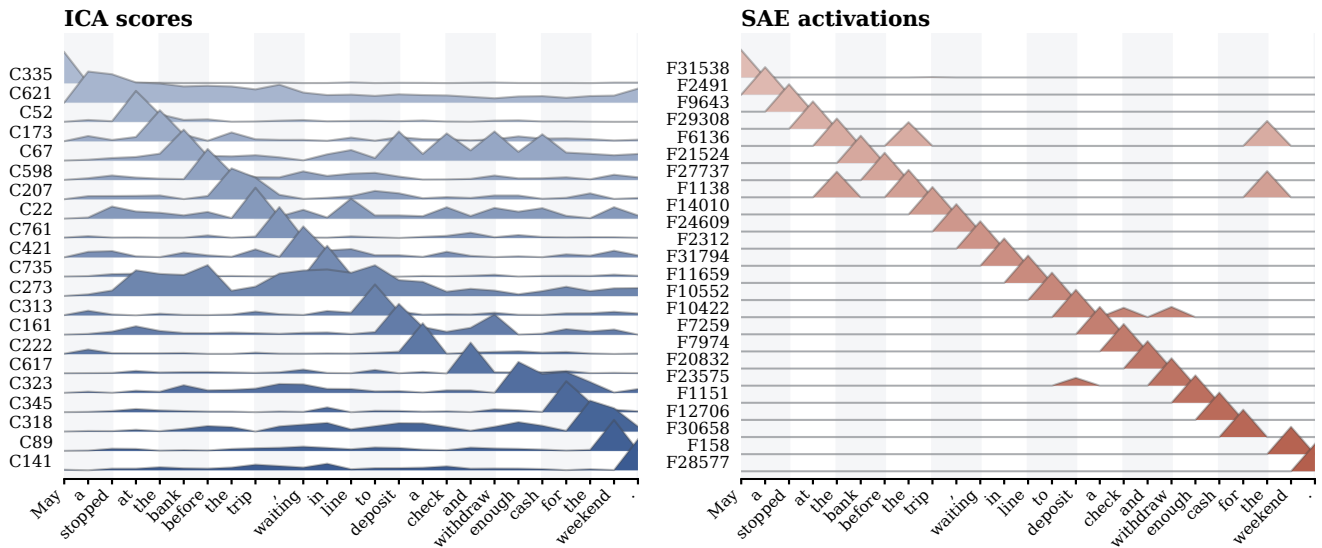


Figure 12: Token-wise ICA and SAE patterns for the same GPT-2 Small sentence. For each token in the bank sentence, we select the top-1 ICA component by absolute score and the top-1 SAE feature by activation, then plot the union of selected directions across all token positions. Each row is normalized by its maximum value in this sentence, so ridge height shows the relative within-direction pattern rather than cross-direction magnitude. The compact ridgeline view highlights a qualitative difference between the dictionaries: SAE features often appear as localized activations, while ICA score directions can trace broader context-sensitive patterns across several related tokens. Sentence: “Maya stopped at the bank before the trip, waiting in line to deposit a check and withdraw enough cash for the weekend.”

Figure 12 shows a qualitative difference between the two lenses. On the SAE side, many selected features appear as localized events, often peaking at one token or a short span. This is consistent with the sparsity pressure in SAE training. On the ICA side, several score directions vary more smoothly across neighboring tokens. Some components remain active over a phrase or local discourse region, such as the financial-bank context around bank, deposit, check, withdraw, and cash. This is the same behavior we saw in the case studies: ICA directions often look less like isolated event detectors and more like compact contextual factors. We observe the same qualitative contrast in Gemma 2 2B and Qwen 3.5 2B Base, with full ridgeline examples provided in Appendix ??.

This difference should not be read as one method being uniformly better. The two decompositions answer different questions. SAE features are optimized to reconstruct activations with sparse feature activations, so localized firing is a desirable outcome of the objective. ICA directions are optimized for non-Gaussianity under a compact basis, so a direction can carry a contextual signal that varies continuously across a span. The ridgeline plot therefore explains why the two methods can be complementary even when their directions partially overlap.

6.3. Feature Utility under Sparse Probing

The previous analyses show that ICA directions are statistically exceptional, human-inspectable, and partially aligned with SAE features. We now ask whether they are useful as feature coordinates. A useful direction should also concentrate task-relevant information in a small number of coordinates.

We test this with SAEBench sparse probing (?). For each concept dataset, SAEBench ranks feature dimensions by how strongly their training-set activations separate positive from negative examples. It then trains a supervised linear probe using only the top- k ranked features. The metric therefore asks whether concept information is concentrated in a small feature budget.

For each input, SAEBench runs the model, collects residual-stream activations at the selected layer, masks special tokens, and averages token-level feature activations into one vector for the whole text. For public SAE baselines, feature activations are obtained from the SAE encoder. For ICA and PCA, feature activations are the corresponding component scores. Since ICA and PCA scores are signed, we split each component into two nonnegative sides,

$$f_i^+ = \max(s_i, 0), \quad f_i^- = \max(-s_i, 0),$$

where s_i is the signed component score. The probe can then select the positive and negative sides separately.

We compare ICA against baselines that cover different notions of feature discovery. Public SAEs provide the strongest high-capacity reference, using large overcomplete dictionaries trained with reconstruction and sparsity objectives. Matryoshka SAEs test whether smaller SAE dictionaries can retain useful features at multiple effective widths (?). ITDA is a training-light sparse-coding alternative that constructs an activation dictionary without gradient-trained SAE latents (?). PCA is a classical linear baseline that captures high-variance directions without optimizing sparsity, non-Gaussianity, or interpretability. These baselines test whether ICA is competitive with public SAEs, whether reducing SAE dictionary size closes the gap, and whether non-Gaussian directions are more useful than standard statistical directions.

All methods are evaluated with the same sparse-probing datasets, feature budgets, and train/test splits. We average over two representative layers for each model: GPT-2 Small layers 6 and 10, Gemma 2 2B layers 12 and 20, and Qwen 3.5 2B Base layers 12 and 20. Since released Matryoshka SAE checkpoints are available for Gemma 2 2B, we compare them only on Gemma 2 2B layer 12.

7. Discussion

7.1. Summary of Findings

This paper revisits ICA as a compact tool for interpreting language-model activations. First, we show that GPU-parallel FastICA, combined with activation normalization and convergence diagnostics, makes layer-wise ICA fitting practical at LLM scale, requiring only minutes per layer in our GPT-2 experiments. Second, ICA directions are not arbitrary rotations: their projection distributions are substantially more

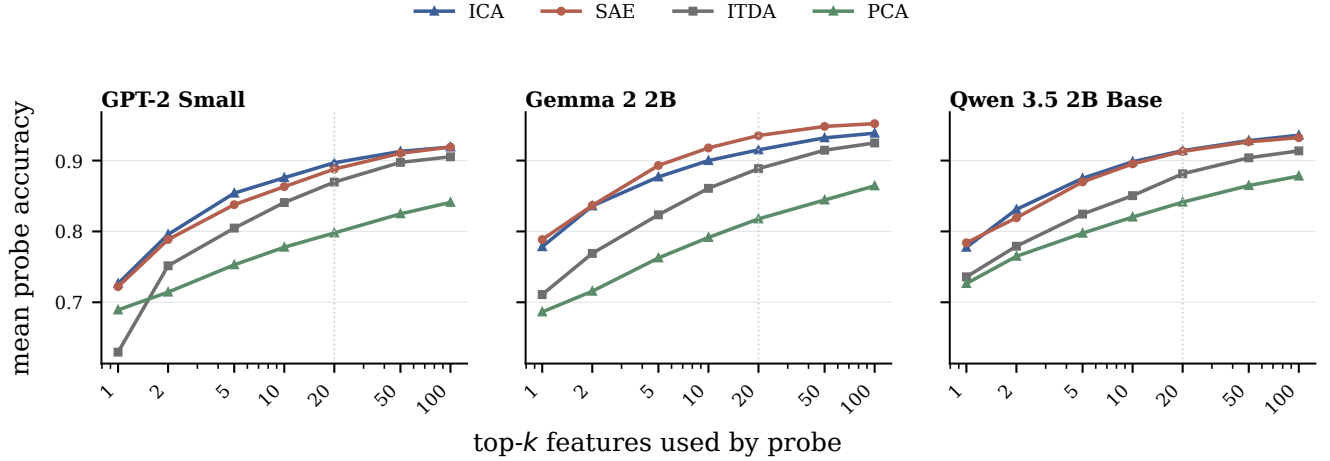


Figure 13: SAEBench sparse-probing performance for ICA, public SAE, ITDA, and PCA representations. For each representation, SAEBench ranks features by class contrast on the training set and trains supervised probes using only the top- k ranked feature activations. Each curve reports test accuracy averaged over the eight default sparse-probing datasets and two evaluated layers for that model. SAE features are produced by the SAE encoder, while ICA and PCA use signed component scores split into two nonnegative sides.

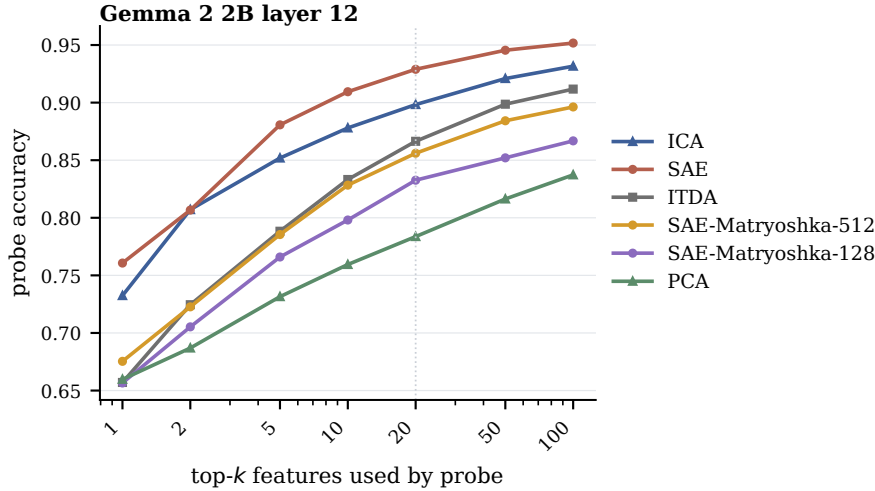


Figure 14: SAEBench sparse-probing comparison for Gemma 2 2B layer 12. We compare PCA, ICA, ITDA, prefix-restricted Matryoshka SAE variants, and the full GemmaScope 16k SAE.

Figure 13 shows that ICA is a strong feature representation across all three model families. ICA consistently outperforms PCA, showing that the gain is not simply due to using a compact linear basis. It also outperforms ITDA and remains competitive with public SAE dictionaries. Figure 14 further tests whether reducing SAE dictionary size closes the gap to ICA on Gemma 2 2B layer 12. Both compact Matryoshka variants stay below ICA across most budgets. Thus, making an SAE dictionary smaller does not by itself make it more competitive with ICA under sparse probing.

non-Gaussian than random directions, and their effective receptive fields reveal a broad spectrum from token-local to context-dependent components. Third, the resulting components support human-readable analysis in both residual-stream and embedding-space case studies, where they decompose polysemy, trace contextual states, and expose lexical structure. Fourth, ICA and SAE directions partially overlap but do not coincide, suggesting that the two methods recover related but distinct views of activation space. Finally, on SAEBench targeted interventions and sparse probing, ICA is competitive in the small-feature regime while avoiding per-layer gradient-based dictionary training.

7.2. ICA as a Complement to SAEs

ICA should not be read as a replacement for sparse autoencoders. SAEs are overcomplete dictionaries trained to reconstruct activations with sparse latent codes. They are designed to build large feature vocabularies and to resolve fine-grained superposition. ICA has a different role. It returns at most a compact basis of d directions, selected by non-Gaussian projection statistics rather than reconstruction loss. This lower capacity limits what ICA can express, but it also makes ICA cheap to fit, easy to browse, and suitable for rapid layer-wise exploration.

A useful workflow is therefore ICA first, SAE when needed. ICA can quickly reveal which layers contain strong non-Gaussian structure, which components are local or contextual, and which directions are promising targets for annotation or intervention. If a component has a close SAE neighbor, the two decompositions provide convergent evidence. If an interpretable ICA component has weak nearest-SAE overlap, it becomes a useful candidate for deeper analysis, since it may reflect structure that is not captured by a single public SAE decoder direction. When the goal is a high-resolution feature dictionary, an SAE remains the more appropriate tool. When the goal is a fast first pass over many layers, models, or activation sites, ICA offers a better cost–interpretability tradeoff.

This positioning also explains the SAE Bench results. Comparing ICA and SAE at the same absolute feature budget is practically meaningful, because human inspection and targeted intervention often operate with small N . It is not, however, a claim of equal capacity. Public SAE dictionaries contain tens of thousands of latents per layer, while ICA exposes a much smaller set of signed directions. The striking result is that, despite this capacity gap and without training an overcomplete dictionary, ICA can still produce strong intended effects in the small- N TPP regime and competitive sparse-probing accuracy. This suggests that some behaviorally useful structure is concentrated in coarse non-Gaussian axes.

Our work also differs from prior ICA baselines in SAE papers. The main change is not a new theoretical objective. It is the combination of a practical GPU-parallel implementation, activation normalization, robust convergence diagnostics, ERF-based context analysis, human inspection, and intervention-based evaluation. Under this lens, ICA is not a failed SAE baseline. It is a lightweight decomposition that occupies a different and useful point in the interpretability toolkit.

7.3. Limitations

The most important limitation is capacity. ICA can recover at most d components from a d -dimensional activation space, or fewer when convergence requires a reduced component count. It cannot provide the large overcomplete vocabulary that modern SAEs are designed to learn. Many concepts may be split across several ICA components, absent from the compact basis, or represented in nonlinear structure that a linear ICA decomposition cannot isolate. A missing or unclear ICA component should therefore not be taken as evidence that the model lacks the corresponding information.

ICA fitting also remains imperfect. Activation normalization and the p95-LIM fallback substantially improve convergence, but some layers still require reduced component counts, and a few layers behave as fitting outliers. When ICA converges at full dimension, the linear transform is close to invertible under the fitted preprocessing, which avoids the learned reconstruction loss of an SAE. When the accepted component count is reduced, however, the pseudoinverse decoder necessarily introduces reconstruction error. Thus, ICA has a useful fidelity advantage in the full-rank setting, but not an unconditional guarantee of lossless reconstruction in every layer.

Our interpretability analysis is also partly qualitative. The random audit and case studies are intended to

demonstrate practical inspectability, not to estimate the true fraction of meaningful directions in a calibrated way. Component labels are working hypotheses, not ground-truth semantic definitions. ERF is an empirical metric with heuristic choices, including evidence-example selection, recovery thresholds, prefix templates, and the maximum tested context length. These choices were useful in practice, but future work should test how sensitive ERF and annotation quality are to each design decision.

The intervention evaluation has its own boundary. SAE Bench was designed for SAE latents, so our TPP experiments require an SAE-compatible wrapper that splits signed ICA scores into positive and negative nonnegative features. This adapter allows a direct comparison, but it is not a native benchmark for ICA. In addition, SAEs become more competitive at larger feature budgets, as expected from their much larger dictionaries. We therefore interpret the intervention results as evidence for ICA’s usefulness in compact low-to-medium budget regimes, not as evidence that ICA should dominate overcomplete SAEs at all scales.

Finally, most of our analysis is performed on residual-stream states. A residual activation contains information accumulated from earlier layers, so ICA on this state tells us what information is present at layer ℓ , but not necessarily what computation layer ℓ newly performs. This distinction matters. A component found in the residual stream may reflect information copied forward from previous layers rather than information created by the current MLP or attention block. Our experiments therefore support ICA as a state-analysis tool, while leaving transformation-level analysis for future work.

7.4. Future Directions

A first direction is to extend ICA beyond residual-stream states. Modern SAE work increasingly studies MLP outputs, attention outputs, transcoders, and skip-transcoders, rather than only residual streams. ICA can follow the same distinction. State ICA would ask what information is present in the hidden state at layer ℓ . Transformation ICA would instead analyze what a layer adds, removes, or amplifies, for example by fitting ICA to MLP outputs, attention outputs, or residual updates. This could separate persistent representation from layer-local computation.

A second direction is hybrid ICA–SAE training. ICA provides stable coarse axes at low cost, while SAEs provide overcomplete refinement. A natural next step is to initialize an SAE decoder or dictionary with ICA components and then train a smaller overcomplete SAE around them. In this workflow, ICA would supply a compact set of high-signal directions, and SAE training would split or refine these directions into finer subfeatures. This may reduce the search cost of SAE training and make the resulting dictionary easier to align with human-interpretable coarse axes.

A related direction is overcomplete ICA. Standard ICA returns a compact basis, but overcomplete ICA variants aim to recover more sources than observed dimensions. If such methods can be made stable at LLM scale, they could bridge part of the capacity gap between ICA and SAEs while preserving ICA’s statistical objective. Even without full overcomplete ICA, deflationary ICA, scalable ICA variants, and better convergence diagnostics may recover additional stable components in layers where parallel FastICA currently requires reduced component counts.

Future work should also expand the evaluation suite. SAE Bench provides a useful external test, but it remains SAE-oriented. Recent work such as Ax Bench and sparse-probing evaluations reports settings where SAEs do not clearly outperform simpler linear baselines in steering, concept detection, or probing (??). These findings make ICA an especially relevant baseline to test. If compact non-Gaussian directions perform well on such tasks, they could help clarify when expensive overcomplete dictionaries are necessary and when simpler linear decompositions are sufficient.

Another important direction is fidelity-centered evaluation. SAE training optimizes sparse reconstruction, yet reconstruction error can still affect model behavior, and scaling studies show that very large dictionaries are needed to reduce such errors (?). ICA provides a useful reference point because a full-dimensional ICA transform is a nearly invertible linear reparameterization of the normalized activation space. This makes it possible to compare SAE reconstruction, ICA reconstruction, and direct activation interventions under output-level metrics such as next-token loss, KL divergence, and downstream task behavior. Such evaluations would help separate interpretability from reconstruction fidelity.

Finally, the annotation pipeline itself can be improved. ERF, top examples, opposite-side examples, trace plots, and context ablations already form a structured evidence set for human labeling. These signals could be turned into a semi-automatic annotation system that proposes labels, tests them with generated prompts, and routes high-ERF components to longer-context analysis. Low-overlap interpretable ICA components are especially valuable targets for such a system, because they may reveal structure that is not immediately visible through existing public SAE dictionaries.

References

A. Preliminaries: Parallel FastICA Pipeline

Independent component analysis (ICA) is a classical method for recovering statistically independent non-Gaussian sources from linear mixtures. Its core intuition is simple: linear mixtures of independent signals tend to look more Gaussian than the original sources. ICA therefore searches for directions whose one-dimensional projections are maximally non-Gaussian (??).

This objective is a natural fit for activation analysis. If a direction in a language model’s activation space is associated with a token form, syntactic pattern, semantic context, or discourse state, then its projection over many token positions may deviate from an ordinary Gaussian projection. ICA gives us a way to search for such statistically exceptional directions without training a separate neural dictionary.

Centering and whitening. Let $X \in \mathbb{R}^{n \times d}$ be an activation matrix, where each row $x_i \in \mathbb{R}^d$ is the activation at one token position. ICA first removes the empirical mean

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i, \quad X_c = X - \mathbf{1}\mu^\top.$$

It then whitens the centered activations so that all retained directions have unit empirical variance and are linearly decorrelated. Let

$$\frac{1}{n} X_c^\top X_c = U \Lambda U^\top$$

be the eigendecomposition of the empirical covariance, restricted to the top $m \leq d$ eigenpairs. The whitening matrix is

$$K = \Lambda^{-1/2} U^\top|_m \in \mathbb{R}^{m \times d},$$

and the whitened data are

$$Z = K X_c^\top \in \mathbb{R}^{m \times n}, \quad \frac{1}{n} Z Z^\top \approx I_m.$$

After whitening, ICA only needs to find an orthogonal unmixing matrix $W \in \mathbb{R}^{m \times m}$. The estimated source scores are $S = WZ$.

FastICA. FastICA finds non-Gaussian projections using a fixed-point update on a smooth approximation to negentropy. We use the standard log-cosh contrast $G(u) = \log \cosh(u)$, with $g(u) = G'(u) = \tanh(u)$ and $g'(u) = G''(u) = \text{sech}^2(u)$. In the parallel variant, all m directions are updated simultaneously:

$$W^+ = \frac{1}{n} g(WZ) Z^\top - \text{diag}(\beta) W, \quad \beta_j = \frac{1}{n} \sum_{k=1}^n g'(w_j^\top z_k). \quad (1)$$

Here g is applied elementwise, w_j is the j -th row of W , and z_k is the k -th whitened activation. After each update, the rows of W^+ are jointly orthonormalized by symmetric decorrelation:

$$W^+ \leftarrow (W^+ W^{+\top})^{-1/2} W^+.$$

This step keeps the components orthogonal and prevents them from collapsing to the same most non-Gaussian direction.

Convergence and scoring. ICA directions are identifiable only up to sign. Standard FastICA therefore measures convergence by the absolute inner product between each direction before and after an update:

$$\text{LIM}_j = \left| \left| \langle w_j^+, w_j \rangle \right| - 1 \right|. \quad (2)$$

A value of zero means that the direction is unchanged up to sign. The standard strict criterion terminates when $\max_j \text{LIM}_j < \tau$, with $\tau = 10^{-4}$ in our experiments.

After fitting, we store the combined ICA projection

$$C = WK \in \mathbb{R}^{m \times d}. \quad (3)$$

For a new activation $\hat{x}$, its signed ICA component scores are

$$s(\hat{x}) = C(\hat{x} - \mu).$$

All later analyses use this fixed scoring map without refitting ICA.

B. Random Annotation Results

B.1. Initial Random Annotation Table

Table 7: Manual labels for the random ICA component audit. Rows are the selected components from `random_components_n50_seed0`. For each component, the table reports only the sign corresponding to its strongest absolute-score example, matching the annotation side most visible in the random-component audit. ERF is the mean effective receptive field measured by the short-context probe.

Layer/Component	Confidence	ERF	Type	Label	Excess kurtosis
GPT-2 Small					
L0/C6	high	1	Word	Arrest, Conviction	46
L0/C192	high	1	Word	After	298
L0/C267	high	1	Word	Any	194
L0/C366	high	1	Word	"	92
L0/C395	high	1	Word	Water, Removed	6.7
L1/C11	high	1	Word	Be, Been	31
L1/C97	high	1	Word	Also, Therefore	15
L1/C109	high	1	Word	\t, \r	23
L2/C302	high	1	Word	]	101
L2/C303	high	1	Form	Letter B	79
L2/C563	high	1	Word	En	80
L3/C257	high	1	Form	Letter Z	49
L3/C291	high	1	Word	Seem, Appear	29
L3/C330	high	1	Word	Very	129
L3/C413	high	1	Phrase	Inverted Is, Are, ...	64
L3/C449	high	1	Form	Letter U	50
L5/C15	high	1	Word	Social Media	26
L5/C696	high	1	Word	Building, Place	15
L6/C392	high	1	Word	Semicolon	92
L7/C140	high	1	Word	Dead, Killed	14
L7/C260	high	1	Word	With	81
L7/C492	high	1	Word	Look, Seem	33
L7/C766	high	1	Word	If	134
L8/C552	high	1	Word	Out, Down, Up	61
L9/C463	high	1	Position	Early Doc Position	11
L9/C621	high	1	Form	Mod-, Sub-	4.3

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Layer/Component	Confidence	ERF	Type	Label	Excess kurtosis
L10/C144	high	1	Form	Letter Z	24
L10/C375	high	1	Word	/,)/	15
L10/C576	high	1	Word	Try, Hard	9.6
L5/C11	high	1.4	Sentence	Sentence start	9.4
L9/C204	high	1.5	Phrase	Than, Comparison	30
L7/C254	high	1.5	Phrase	Thus, Therefore starting a clause	11
L4/C298	high	1.8	Phrase	Ongoing -ing	18
L6/C290	high	1.8	Global	Geopolitical conflicts	7.8
L4/C8	high	1.9	Form	D-	8.6
L2/C152	high	2	Phrase	Code whitespace	19
L7/C17	high	2	Sentence	Scientific research, Citation	33
L10/C74	high	2	Form	XML tag name	81
L5/C585	high	2.1	Phrase	i.e., Mean	12
L6/C164	high	2.1	Sentence	Citation	17
L8/C736	high	2.3	Phrase	Right, Permission, License to	3.0
L9/C118	high	3.1	Sentence	If-condition	7.9
L10/C368	high	3.2	Global	Gaming language	31
L9/C445	high	5.3	Long-Range Context	Refer to a spokesperson	24
L3/C294	medium	1	Word	Happy, Able	13
L6/C246	medium	2.2	Phrase	in, the, or to following tell or say	9.7
L10/C146	medium	5.4	Global	Natural area	11
L8/C738	medium	7.7	Long-Range Context	Repetition of a prior xxx: section header	23
L10/C142	low	8.3	Sentence	Conditional repetition	18
L2/C648	unclear	11+		?	-1.1
Gemma 2 2B					
emb/C1964	high	1	Word	kick	13
emb/C2034	high	1	Word	Smiley emoji	89
L0/C9	high	1	Word	Have	163
L1/C82	high	1	Word	= " in code	124
L2/C302	high	1	Word	Does, Is, Will	134
L2/C341	high	1	Phrase	Two leading whitespaces	91
L2/C708	high	1	Word	Array	25
L2/C890	high	1	Word	Data	234
L5/C404	high	1	Word	Non-breaking space, U+00A0	25
L7/C351	high	1	Word	Do, Did	80
L16/C495	high	1	Word	Get	4.4
L19/C171	high	1	Form	Letter X	60
L20/C79	high	1	Word	Change	13
L20/C183	high	1	Word	Out	24
L5/C602	high	1	Word	Move	14
L9/C442	high	1	Word	Based, In response, Depend	7.1
L21/C399	high	1.1	Phrase	Adj. prefer after be	17
L7/C530	high	1.1	Phrase	Heart, Liver, Lung	27
L9/C79	high	1.1	Word	Director, President	21
L24/C355	high	1.2	Global	Turkish language	28
L21/C575	high	1.2	Phrase	Opening parenthesis in code	6.8
L15/C65	high	1.3	Phrase	Anatomical location	8.6
L10/C311	high	1.4	Phrase	Be the, Be a	23
L19/C210	high	1.6	Form	Letter V	12
L16/C527	high	1.8	Phrase	Family name	35
L18/C19	high	2.2	Sentence	Gas, Fluid	6.5
L6/C541	high	2.2	Phrase	What, How question	18
L16/C206	high	2.2	Global	Low-level programming language, C, ASM, ...	13
L6/C677	high	3.5	Sentence	If-condition	17
L9/C516	high	4.2	Sentence	Polynomial, Quadratic expression	2.6
L20/C263	high	4.4	Global	Cryptography, Computer Security	5.4
L22/C218	high	4.4	Sentence	Persistent deficiency despite expected improvement	5.9
L12/C466	high	4.6	Global	Accusation of bad behavior	4.3
L24/C372	high	5.2	Global	Immigration policy discussion	6.3
L18/C206	high	5.7	Long-Range Context	Same line ending in code	31
L8/C303	high	6	Sentence	At/In/During + time, something happened/changed	1.2
L19/C543	high	6.2	Sentence	Expecting additional parallel elements	29
L24/C510	high	6.6	Global	Female-centered narrative	10

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Layer/Component	Confidence	ERF	Type	Label	Excess kurtosis
L25/C854	high	7.5	Phrase	Citation marker	94
L23/C567	high	7.7	Global	UK civic info	6.8
L13/C30	high	8.6	Sophisticated	Refer back to one of two recently introduced alternatives	9.8
L10/C261	high	9.2	Long-Range Context	Repetition and number increases	35
L6/C775	high	11+	Long-Range Context	Repetition of prefix	19
L7/C106	medium	1.1	Sentence	Beginning of a new paragraph	7.7
L20/C101	medium	1.6	Phrase	Number >10	40
L25/C92	medium	2.9	Global	Core issue in news	16
L6/C216	low	3	Phrase	71 in No.	0.5
L5/C453	low	3.5	Form	To-infinitive	0.4
emb/C59	unclear	1.7		?	0.2
L18/C33	unclear	11+	Long-Range Context	Apostrophe with unknown pattern	75
Qwen 3.5 2B Base					
emb/C930	high	1	Word	Where	107
emb/C984	high	1	Word	Sort	24
L0/C446	high	1	Word	Control	43
L1/C694	high	1	Word	Google, Big Tech Company	164
L1/C917	high	1	Word	Comma + a/the	152
L1/C1096	high	1	Word	Down	191
L1/C1604	high	1	Word	As (Capitalized)	186
L2/C511	high	1	Word	Shoot	138
L4/C1591	high	1	Word	Advertise	26
L7/C730	high	1	Position	Post-title article start	12
L7/C1009	high	1	Word	Without	35
L13/C1634	high	1	Word	Create, Develop	51
L16/C1572	high	1	Word	Only	150
L16/C1811	high	1	Form	Word fragment -e	24
L19/C808	high	1	Word	Chinese atmospheric literary narrative character	17
L21/C519	high	1	Word	>	49
L3/C508	high	1.1	Word	Register	14
L2/C1940	high	1.1	Phrase	End	268
L6/C647	high	1.3	Phrase	Last as in "last year/week"	44
L20/C189	high	1.3	Phrase	Total number, Amount	17
L13/C3	high	1.3	Phrase	Sensitive identity	14
L4/C1899	high	1.9	Word	Money, Fund	95
L13/C392	high	1.9	Sentence	Legal document format	56
L8/C522	high	2	Phrase	Is, Was, Has	32
L19/C824	high	2	Phrase	Follow, violate rules	23
L19/C999	high	2.4	Phrase	Math quantity, Math object	15
L11/C1211	high	2.4	Phrase	Because of, As a result of	13
L3/C77	high	2.5	Phrase	I feel I, You feel you, Feel like, ...	5.9
L13/C577	high	2.5	Sentence	"In" used in software context	10
L18/C280	high	3.1	Global	Reproductive health	20
L18/C1794	high	3.5	Global	Study protocol	46
L20/C970	high	4	Sentence	Motor, Electromechanical device	15
L18/C745	high	4.3	Global	Object in mechanical patent	21
L15/C258	high	4.4	Phrase	Age of a person or group	30
L22/C511	high	4.9	Sentence	Either or, Whether or, One way or another	5.0
L21/C145	high	4.9	Phrase	Citation (pandoc-style with dashes inside the ID)	29
L14/C707	high	5.3	Phrase	Legal case citation	16
L8/C189	high	5.3	Sentence	Number in elementary arithmetic	25
L9/C47	high	5.6	Sentence	Contrast, Opposition	12
L19/C526	high	8	Global	Javascript i18n code	17
L7/C671	medium	6.1	Phrase	Closing bracket in code	20
L1/C834	low	3.6	Phrase	Certain name	1.0
L13/C1679	low	4	Sentence	Context where medical condition or biological process happens	7.4
L6/C441	low	11+	Long-Range Context	Beginning of a repeated labeled/enumerated entry after intervening context	3.4
L23/C1250	low	11+	Sentence	Memorized string in code snippet	184
L7/C406	unclear	4.6		?	0.5
L23/C541	unclear	5.6		?	1.2
L5/C126	unclear	7.4		?	0.5

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Layer/Component	Confidence	ERF	Type	Label	Excess kurtosis
L3/C1037	unclear	7.5		?	0.5
L19/C646	unclear	10.3		?	11

B.2. Secondary Contrastive Label Audit

To reduce the risk that the high-confidence labels reflect only one annotator’s idiosyncratic reading of top examples, we performed a secondary contrastive audit of the high-confidence subset of the random annotation study. This audit covered all 127 components whose initial annotation confidence was marked "high": 44 from GPT-2 Small, 43 from Gemma 2 2B, and 40 from Qwen 3.5 2B Base.

The audit protocol was as follows. For each component, the auditor first inspected the proposed label together with representative top-scoring examples from the database. They then wrote matched contrastive prompts: label-consistent prompts intended to activate the component, and label-inconsistent controls that reused similar surface forms when possible. The frozen model and fitted ICA decomposition were run on these prompts. Because autoregressive models only condition on left context, target tokens were chosen so that the relevant evidence appeared before or at the target position. For labels assigned to the negative sign of an ICA component, scores were sign-flipped so that larger signed scores always indicated stronger evidence for the proposed label. The audit decision used both the signed score at the intended target token and the largest signed component score anywhere in the prompt, producing a 0–10 audit score and a suggested refined label phrase.

The audit supported 121 of the 127 high-confidence labels and partially supported the remaining 6; none were judged unsupported. On a 10-point audit scale, the score distribution was: one label scored 5, two scored 6, twelve scored 7, thirty-five scored 8, forty-one scored 9, and thirty-six scored 10. Thus 112 of 127 high-confidence labels scored at least 8. The partially supported cases typically involved labels that were directionally correct but too broad, too narrow, or sensitive to a specific token site, such as markup punctuation or context-dependent function words. Table 8 summarizes the initial and suggested labels for the audited subset.

Table 8: Secondary audit summary for the high-confidence audited components from the random ICA component annotation study. Rows are drawn from `random_components_n50_seed0`. Score is the 0–10 secondary audit score.

Layer/Component	Initial label	Score	Suggested label
GPT-2 Small			
L0/C6	Arrest, Conviction	9	arrest / conviction / guilty legal context
L0/C192	After	10	after token
L0/C267	Any	10	any token
L0/C366	"	8	double quote token
L0/C395	Water, Removed	8	water token
L1/C11	Be, Been	9	be / been token
L1/C97	Also, Therefore	8	also / therefore discourse connective
L1/C109	\t, \r	10	tab / carriage-return token
L2/C302	}	10	closing square bracket token
L2/C303	Letter B	10	B / b token
L2/C563	En	8	en subword token
L3/C257	Letter Z	10	Z / z token
L3/C291	Seem, Appear	9	seem / appear token
L3/C330	Very	10	very token
L3/C413	Inverted Is, Are, ...	8	inverted/question auxiliary
L3/C449	Letter U	10	U / u token

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Layer/Component	Initial label	Score	Suggested label
L5/C15	Social Media	9	social-media platform/context
L5/C696	Building, Place	8	building / apartment place token
L6/C392	Semicolon	10	semicolon token
L7/C140	Dead, Killed	9	died / killed / dead token
L7/C260	With	10	with token
L7/C492	Look, Seem	9	look / seem / sound evidential verb
L7/C766	If	10	if token
L8/C552	Out, Down, Up	9	out / down / up particle
L9/C463	Early Doc Position	8	early-document opening subtoken
L9/C621	Mod-, Sub-	8	mod/sub prefix token
L10/C144	Letter Z	10	Z / z token
L10/C375	/,)/	8	slash token in URL/path/markup
L10/C576	Try, Hard	8	try / trying / tries token
L5/C11	Sentence start	9	sentence-initial capitalized token
L9/C204	Than, Comparison	10	than comparison token
L7/C254	Thus, Therefore starting a clause	9	thus / therefore consequence connective
L4/C298	Ongoing -ing	9	progressive/ongoing -ing form
L6/C290	Geopolitical conflicts	9	geopolitical conflict/diplomacy context
L4/C8	D-	9	D-initial subword/token
L2/C152	Code whitespace	7	code indentation/alignment whitespace
L7/C17	Scientific research, Citation	9	scientific reference-title final period
L10/C74	XML tag name	10	HTML/XML closing tag name
L5/C585	i.e., Mean	8	i.e. / that is to say / mean
L6/C164	Citation	8	reference-list title-final period
L8/C736	Right, Permission, License to	9	permission/right/license to
L9/C118	If-condition	8	then/comma after if-condition
L10/C368	Gaming language	9	Video-game discourse
L9/C445	Refer to a spokesperson	8	spokesperson/spokesman name reference
Gemma 2 2B			
emb/C1964	kick	10	Kick / kicks token
emb/C2034	Smiley emoji	7	Emoji / emoticon token
L0/C9	Have	8	have / has / had auxiliary
L1/C82	" in code	8	Quoted attribute assignment in markup
L2/C302	Does, Is, Will	10	WH-question auxiliary does/is/will
L2/C341	Two leading whitespaces	10	Two-space indentation token
L2/C708	Array	9	array token
L2/C890	Data	10	data token
L5/C404	Non-breaking space, U+00A0	10	Non-breaking space (U+00A0)
L7/C351	Do, Did	10	do / did auxiliary
L16/C495	Get	8	get / got / gotten
L19/C171	Letter X	10	X / x token
L20/C79	Change	9	change / changes / changed
L20/C183	Out	9	out token
L5/C602	Move	8	move / moved / moving
L9/C442	Based, In response, Depend	9	based / in response / depend construction
L21/C399	Adj. prefer after be	7	predicate adjective after be
L7/C530	Heart, Liver, Lung	7	cardiac / cardiovascular and organ terms
L9/C79	Director, President	6	institutional role title
L24/C355	Turkish language	10	Turkish-language token/subword
L21/C575	Opening parenthesis in code	8	opening call/index delimiter in code
L15/C65	Anatomical location	9	anatomical term/subword
L10/C311	Be the, Be a	9	article after be
L19/C210	Letter V	10	V / v token
L16/C527	Family name	10	surname after given name/title
L18/C19	Gas, Fluid	8	gas/fluid flow medium
L6/C541	What, How question	9	What/How question auxiliary
L16/C206	Low-level programming language, C, ASM, ...	8	low-level systems-code token
L6/C677	If-condition	9	If-clause predicate after if this/it/that

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Layer/Component	Initial label	Score	Suggested label
L9/C516	Polynomial, Quadratic expression	8	minus sign in polynomial/algebra expression
L20/C263	Cryptography, Computer Security	8	cryptography / security context
L22/C218	Persistent deficiency despite expected improvement	7	still/remain poor despite expectation
L12/C466	Accusation of bad behavior	8	accusatory / blame discourse
L24/C372	Immigration policy discussion	8	immigration-policy news discourse
L18/C206	Same line ending in code	7	repeated code/list line-ending pattern
L8/C303	At/In/During + time, something happened/changed	7	temporal transition after time expression
L19/C543	Expecting additional parallel elements	7	parallel/list element expectation
L24/C510	Female-centered narrative	8	female-character narrative context
L25/C854	Citation marker	6	pandoc-style citation marker digit
L23/C567	UK civic info	7	UK public-service/civic information prose
L13/C30	Refer back to one of two recently introduced alternatives	9	reference to one member of a recent pair
L10/C261	Repetition and number increases	8	repeated identifier with increasing numeric suffix
L6/C775	Repetition of prefix	8	repeated identifier prefix
Qwen 3.5 2B Base			
emb/C930	Where	10	where / Where token
emb/C984	Sort	10	sort / Sort token
L0/C446	Control	10	control token
L1/C694	Google, Big Tech Company	9	Google token / big-tech context
L1/C917	Comma + a/the	9	article after comma appositive
L1/C1096	Down	10	down token
L1/C1604	As (Capitalized)	10	capitalized As token
L2/C511	Shoot	10	shot / shoot token
L4/C1591	Advertise	9	advertising / ads token
L7/C730	Post-title article start	9	article-start token after scientific title
L7/C1009	Without	10	without token
L13/C1634	Create, Develop	9	create/develop verb
L16/C1572	Only	10	only token
L16/C1811	Word fragment -e	9	word-final e fragment
L19/C808	Chinese atmospheric literary narrative character	8	Chinese literary/poetic atmosphere character
L21/C519	>	8	literal > in markup/code
L3/C508	Register	10	register token
L2/C1940	End	10	end / End token
L6/C647	Last as in "last year/week"	9	last in temporal expression
L20/C189	Total number, Amount	9	aggregate count/amount expression
L13/C3	Sensitive identity	9	sensitive identity term
L4/C1899	Money, Fund	9	money / funds token
L13/C392	Legal document format	8	legal-document header formatting
L8/C522	Is, Was, Has	5	is/was after discourse setup
L19/C824	Follow, violate rules	10	rule compliance/violation language
L19/C999	Math quantity, Math object	8	scientific/mathematical quantity term
L11/C1211	Because of, As a result of	9	causal preposition phrase
L3/C77	I feel I, You feel you, Feel like, ...	8	pronoun after feel/felt like
L13/C577	"In" used in software context	8	in within code/software context
L18/C280	Reproductive health	7	reproductive/adolescent health context
L18/C1794	Study protocol	10	clinical study protocol/approval language
L20/C970	Motor, Electromechanical device	9	electromechanical motor/device context
L18/C745	Object in mechanical patent	9	mechanical patent object term
L15/C258	Age of a person or group	9	age number for person/group
L22/C511	Either or, Whether or, One way or another	9	alternative/concessive construction
L21/C145	Citation (pandoc-style with dashes inside the ID)	9	dash inside Pandoc citation/key
L14/C707	Legal case citation	7	legal case citation whitespace
L8/C189	Number in elementary arithmetic	9	digit answer/context in elementary arithmetic

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Layer/Component	Initial label	Score	Suggested label
L9/C47	Contrast, Opposition	8	contrast continuation token
L19/C526	Javascript i18n code	7	JavaScript/JSON locale object